Heitaria Swiss AG
ASSET MANAGEMENT

TECHNICAL REPORT · PHASE 0

SALAR DE UYUNI — NOR LÍPEZ, POTOSÍ

Energy Supply Assessment

Phase 0 — desk assessment of the techno-economic model

PROJECT	Lithium carbonate, 80,000 t/y plus a 10,000 t/y pilot plant, from brine of the lithium-rich zone of Nor Lípez. Route C process flow diagram, with no evaporation ponds.
SUBJECT	The plant's energy supply: 49.5 MW of continuous electrical load and 119.9 MW-t of thermal load — the finishing heat plus the steam that evaporates the brine — supplied on a self-sufficient basis — off-grid — by a 283.6 MW photovoltaic field, 792 MWh of batteries, a 389.1 MW-t solar thermal field with molten-salt storage, and a 21.2 MW diesel firm-supply reserve dedicated exclusively to safe hold.
TECHNICAL BASIS	The project's techno-economic model in its v22 version as at 28 August 2026 (evaporation basis, Power, Notes, Firm Supply (Diesel), Heat (CSP), GEA & Grade, Opex Build-Up and Dashboard tabs), the site master plan, and the third-party sources the model itself cites. No on-site meteorological or performance data have been obtained for this project.
ADDRESSEES	The project's lenders and export credit insurer — Euler Hermes / UFK and the entities of the financing sub-group —; CUPCONL, Central Única Provincial de Comunidades Originarias de Nor Lípez (the provincial umbrella body of the Indigenous communities of Nor Lípez); and the technical advisers appointed by either of them.

Version 2.2 — 7 October 2026. Adds Annex A, on what European battery buyers will ask about energy and carbon under the EU Batteries Regulation. Sections 6.2 and 6.5 now take the water surplus from the Hydrological and Hydrogeological Assessment (8,165 m³/day); no change to the energy figures. Version 2.1 — 9 September 2026.

This is a planning-grade desk assessment. It is not an engineering study and does not carry the signature of a registered professional. All cost figures are Class 5, with a declared band of ±40%. No site-specific typical meteorological year is available, nor an hourly load profile for the plant, nor vendor-certified derating curves; all three are first-order inputs and their absence is flagged in every section where it weighs. Section 8 sets out the programme that will replace them.

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RESPONSES AND ENQUIRIES

Observations from appointed advisers may be addressed in writing to this address and will be answered in writing.

EXECUTIVE SUMMARY

The standing of this assessment

Every figure in this document comes from the project's own techno-economic model, is derived arithmetically from it, or comes from the third-party sources the model cites. No site measurements have been taken for this project: no campaign to measure the solar resource, temperature or soiling has yet been carried out.

That limits what can be concluded, and here it limits more than it did in the case of water. The volume of water is fixed by production, grade and recovery, all on paper. Energy is not: the size of the photovoltaic field depends on a capacity factor that only local measurement can establish, and that of the firm-supply reserve depends on the actual frequency of snowfall episodes, for which no site record has yet been obtained. What can be done — and it is what this document does — is to run the arithmetic across the entire plausible range of both and show where the answer is insensitive to it, because there it constitutes a result and not an assumption.

Purpose of this copy. It is delivered simultaneously to two distinct readerships and has been written to serve both without changing figures from one to the other. For the lender and the credit insurer, it is the support for the firm-supply case: what was bought, what it protects, how many hours a year the fossil backup runs and what that does to debt service coverage. For CUPCONL, it is the statement of what the energy system takes from the territory and what it does not: it takes no energy from the regional grid, it hauls no fuel along the route that floods from December to March, it uses no fresh water to clean the panels, and it occupies on the order of seven square kilometres whose location is a matter for the master plan.

It is delivered in the same form in which it is used internally. It carries no request for a decision.

49.5 MW	119.9 MW-t	283.6 MW	206 h/y
CONTINUOUS ELECTRICAL LOAD	THERMAL LOAD	PHOTOVOLTAIC FIELD	RUNNING HOURS OF THE DIESEL GENSETS

Electrical load: 45.0 MW of plant and auxiliaries plus 4.5 MW for the mechanical vapour recompression compressor. Thermal load: 39.8 MW-t of finishing heat plus 80.2 MW-t of evaporation steam. Photovoltaic field at a 23% capacity factor; at P90 — 21% — it would be 310.6 MW. Sections 2 and 3.

CRITICAL SYNTHESIS

The most important results

There is no grid, and there would be none even if the line were built. The site has no reliable electricity supply. Nor could the national system carry this load: two thirds of Bolivian generation is gas-fired, domestic gas production has fallen by more than 56% since 2015, and this one plant is equivalent to 9.7% of the country's peak demand in total energy, 2.8% in electricity alone. Building a dedicated supply is therefore not an environmental preference. It is the only option available. And because there is no grid to fall back on, the system cannot be sized for an average day. It is sized for the worst event likely within the life of the asset: three to four days of dense cloud or snowfall. A system sized on the annual average would look adequate on paper and fail in January.

The plant runs mostly on heat, not on electricity. The brine is not boiled with electricity. It is evaporated with solar steam in closed crystallisers, and mechanical recompression reuses each tonne of that steam 3.5 times, which cuts the quantity of steam required. The plant therefore needs 49.5 MW of electricity and 119.9 MW-t of heat, supplied by a 283.6 MW photovoltaic field and a 389.1 MW-t solar thermal field. Two thirds of the energy is heat, and heat is the cheaper two thirds: 63 dollars per MWh against 139.

The energy system is expensive to build and cheap to run. It costs about USD 1,107 million to build — 32% of the USD 3,465.4 million project — and then USD 157 per tonne to operate, which is 3.2% of operating cost. That is the shape being financed: the risk sits in construction and in a ±40% estimate, not in a fuel price over forty-five years.

In a storm the plant is protected, not kept in production. When several days pass without sun, production stops. The plant sheds 76% of its load and drops to about 12 MW — enough to keep brine circulating, lines above crystallisation temperature and the plant safe. This condition is called safe hold, and it is what the backup is designed for. At that reduced load the batteries last 44.5 hours and the molten salt 112.4 hours, so the heat is covered for the whole of a four-day event. Diesel generators cover only the electricity the batteries cannot reach: 51.5 of the 96 hours. That is why the backup is 21.2 MW and USD 18.7 million, and not the 117 MW and USD 107 million that keeping the plant in production would have cost.

One input still has to be confirmed. The design solar irradiance — 2,700 kWh/m² per year — is taken from published data for the region and has not been measured on site, and it sets the size of the solar thermal field, the largest capital item in the system. A measurement station settles it within a year — Section 8.2.

DESCRIPTION

The project

The project is a lithium carbonate plant on the Salar de Uyuni, in Nor Lípez, Potosí. It will produce **80,000 tonnes a year**, preceded by a 10,000 t/y pilot, from brine drawn out of YLB's existing wellfield. It uses **no evaporation ponds**: the brine is concentrated in closed crystallisers instead. That single decision is why the plant is energy-intensive and why it produces more water than it consumes. It costs **USD 3,465.4 million** to build, operates for 45 years and employs 702 people permanently. The energy supply — the subject of the rest of this document — is USD 1,107 million of that cost and 169 MW of continuous demand.

This page is description, not assessment. Its figures come from parts of the model other than those supporting the rest of the document; they are identified in Section 9.1, and where they are uncertain this is stated.

ITEM	VALUE
Product	Lithium carbonate (Li_2CO_3).
Steady-state production	80,000 t/y, plus a 10,000 t/y pilot plant that precedes it.
Resource	Brine from the lithium-rich zone of Nor Lípez: 12.23 Mm^3/y abstracted from the existing YLB (Yacimientos de Litio Bolivianos, the state lithium company) wellfield, at a design grade of 1.5 g/L lithium and 82% recovery. The project opens no new wellfield.
Process route	Route C, **with no evaporation ponds**: evaporation in closed crystallisers driven by solar steam, with mechanical recompression that reuses each tonne of steam 3.5 times. It is this decision that makes the project energy-intensive and, at the same time, a net producer of water.
Phases	Pilot built in year 1, tested in year 2 and commercial from year 3. Full plant built between years 3 and 5, commercial in year 6.
Modelled life	45 years, of which 43 in production.
Total cost	USD 3,465.4 million, of which USD 1,106.7 million corresponds to the energy system this document assesses.
Structure	Bolivia 51% and sponsor 49%. The sponsor contributes 100% of the plant's equity; Bolivia contributes no cash.
What it takes from YLB	A single item: the use of the salt disposal cells, at no cost. Brine, water and energy are self-supplied.

Employment and presence in the territory

PERMANENT POSITIONS IN OPERATION	PEOPLE	CONTENT
Operations	390	Continuous 80,000 t/y plant, four-shift roster.
Maintenance	187	Mechanical, electrical and instrumentation.
Technical staff, laboratory and administration	125	Metallurgy, quality control, health and safety, finance.

PERMANENT POSITIONS IN OPERATION	PEOPLE	CONTENT
Total permanent in operation	**702**	**Rotational regime, with some 12 rotations a year and transport to Uyuni and Oruro.**
Construction peak	**≈1,400**	**Estimate of the workforce at the peak of construction, between years 1 and 5. It is the basis on which the residential camp and its water plant were sized.**

The two figures describe different phases and must not be added together. The construction peak is transitory and larger; the 702 positions are permanent and are the ones that remain through the 43 years of production. Sizing the camp and its water plant against the construction peak, and not against the steady-state complement, is the correct practice: the facility is built once, for the moment of greatest occupancy.

TECHNICAL MATRIX

Key questions and preliminary answers

QUESTION	ANSWER AT THIS STAGE	BASIS
Does the project take energy from the national grid or from YLB?	No. None, in any operating case. There is no reliable grid at the site, and the national system could not absorb it either: the plant is equivalent to 9.7% of the Bolivian peak in total energy and 2.8% in its electrical part alone. Nor is there any tariff at YLB cost.	Section 1.1. The only thing taken from YLB is the salt disposal cells, at no cost.
How much energy does the plant consume?	49.5 MW electrical continuous — 433.4 GWh a year — and 119.9 MW-t of thermal load — 1,050.6 GWh-t a year — of which 80.2 MW-t is the steam that evaporates the brine. That is 18,550 kWh per tonne of carbonate.	Section 2. Derived from the model; the hourly profile is not yet documented.
Where does that energy come from?	From 283.6 MW of fixed-tilt photovoltaics, 792 MWh of batteries with 16 hours of autonomy, and a 389.1 MW-t solar thermal field with 15 hours of molten-salt storage.	Section 3.
What happens when there is no sun for several days?	Production is shed and the plant goes into safe hold, some 12 MW. The batteries sustain 44.5 hours on their own and the molten salt covers the hold heat for the whole of the event; beyond that, seven 3.5 MW diesel gensets come in, which protect the plant, not production.	Section 4. The 12 MW hold load is an engineering estimate, not a process balance figure.
How much diesel does the project burn?	**In emergency only:** 618,000 litres a year, in some 206 hours of running. The gensets do not generate in any normal operation. The 232.0 million litres a year of a wholly diesel supply are a counterfactual: they are not bought.	Sections 3.4, 4.4 and 5.3.
Will tanker trucks be running through the region?	The backup requires a 772 m³ tank farm, filled once per season. An equivalent diesel plant would have required on the order of twenty-two tankers a day, along a route that floods from December to March.	Section 6.4.
How much water is needed to clean the panels?	Between 18 and 24 cleanings a year with low-mineralisation water, taken from the surplus process condensate. No fresh water is abstracted. The volume per cleaning is not yet documented.	Section 6.2 and Hydrogeological Assessment §5.4.
How much land does it occupy?	4.73 km² for the photovoltaic field with fixed-tilt structure at normal density, against 6.25 km² allocated: a third is left over, and the P90 case also fits.	Section 6.1.
What emissions does it avoid?	619,978 t of CO_2 a year relative to making the same energy with diesel, net of what the backup actually burns. Over 45 years, some 27.9 million tonnes.	Section 6.3. Arithmetic on the model's emission factor.
How much does it cost and what weight does it carry in the project?	USD 1,106.7 million of capital — 32% of the total cost — and USD 157.2 per tonne in named line items, 3.2% of the operating cost. Both figures with a ±40% band.	Section 5.
Is it cheaper than diesel?	Yes, and by more than before on the heat side. The levelised cost of the self-supplied electricity is 139 $/MWh against 438 $/MWh for diesel fuel alone; that of the solar heat is 63 $/MWh-t against 206 $/MWh-t.	Section 5.3.
Do the batteries last the 45 years of the project?	No. A calendar life of 15 years and two cell replacements are planned, in years 20 and 35 of the project, for USD 199.2 million nominal. That calculation still omits the battery of the recompression compressor.	Section 5.4.
Were nuclear or wind power considered?	Microreactors were assessed and discarded as a design option: there is no commercial licence, no deployed reactor and no published price. Wind has not been measured at the site and the regional literature points against it: a poor resource and a high season from June to October, the opposite of the design event from January to March.	Sections 1.3 and 3.5.
Which inputs move these figures the most?	Three, in order. The design direct normal irradiance of 2,700 kWh/m² a year sets the size of the solar thermal field, the largest capital item in the system. Next, the specific consumption of the recompression compressor, which sets the electrical load. Third, the photovoltaic capacity factor: at P90 — 21% — the field grows to 310.6 MW and the capital by some USD 28 million.	Sections 1.2 and 3.3.

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Convention for figures. All amounts are in United States dollars. Capital costs are Class 5 with a ±40% band unless expressly stated otherwise. Operating figures are expressed per tonne of lithium carbonate at the steady-state production of 80,000 t/y. “MW” denotes electrical power and “MW-t” thermal power; “GWh” electrical energy and “GWh-t” thermal energy. Totals are calculated on unrounded values, so a column may not add exactly to the total heading it. Where this document differs from a figure recalled from an earlier reading, the figure in this document is the one the model produces today.

SECTION 01

1. Physical and energy framework

1.1 The site has no grid

There is no reliable electricity supply at the site. Buying power from YLB over the Potosí grid was examined and ruled out: there is no dependable grid to buy it from, and no tariff to buy it on.

It is worth being explicit about why this matters more than it appears, because the absence of a grid is not simply a cost item that moves from operations to capital. It changes three things fundamentally.

- **Instantaneous backup disappears.** A connected plant can size its own generation against the mean and let the grid cover the gaps. Here every kilowatt-hour must be produced at the instant it is consumed or have been stored beforehand. There is no one to buy from at three in the morning on a cloudy day.
- **The design case ceases to be the average day.** It becomes the worst event with a reasonable probability of occurring within the life of the asset. The whole of Section 4 exists for this reason, and it is the section a lender should read first.
- **There is no tariff at cost, and none is assumed.** The project self-supplies brine, water and energy. The only thing it takes from YLB, at no cost, is the use of the salt disposal cells, and the operating cost is built from the bottom up on that basis.

For the community reading of this document, the consequence is direct and it is favourable: the project does not compete for energy with the regional supply, does not connect to it, and does not depend on its being reinforced. For the lender's reading, the consequence is the opposite and must also be stated: there is no grid to absorb a sizing error, and that is why the reserve margin of Section 3 is not optional.

Why there would be no grid to draw on either

The above is a fact about the site: there is no line. The fact about the system should be added, because it changes the nature of the argument. Even if the line existed, the Sistema Interconectado Nacional (SIN, the national interconnected system) is today in no condition to absorb a load of this size. The official 2024 figures are as follows.

SIN INDICATOR — OFFICIAL 2024 DATA	VALUE	NOTE
Installed capacity	3,787 MW	Nominal. Not all of it usable when gas is scarce.
Effective available capacity	3,318 MW	Effective fleet as at December 2024.
Peak demand	1,752 MW	Record, 25 September 2024.
Total annual generation	11,689 GWh	+3.8% on 2023.
Share of gas in generation	65 to 70%	Hydro ~28%; wind and solar, a low share.

On paper the headroom looks ample: 3,787 MW installed against a 1,752 MW peak. In practice that margin is limited by fuel and not by hardware. Between two thirds and seven tenths of national generation depends on natural gas, domestic gas production has fallen by more than 56% from its 2015 maximum, Bolivia has been a net importer of fuels since 2022, and blackout warnings have been issued for the SIN for 2025 and 2026. Potosí shares that grid and inherits that fragility.

SCALE

This plant, measured against the country

The consolidated energy demand of Section 2 — 169.4 MW continuous and 1,484.0 GWh a year — is equivalent to **9.7% of Bolivia's national peak demand and 12.7% of its entire annual electricity generation**. Even counting only the electrical part, which is what a grid would have to deliver, it is 49.5 MW continuous: 2.8% of the national peak and 3.7% of annual generation.

The two readings pull apart, and this should be said. The *electrical* load is modest — 49.5 MW, so the part a grid would have to deliver is a quarter of the total. The *total* load is large, because the steam that evaporates the brine costs some 179 thermal kWh per cubic metre of water where fully electric recompression would cost 35 electrical kWh. It is more abundant and much cheaper energy per unit — 63 dollars per thermal MWh against 139 per electrical MWh — but it is more energy, and that is the correct way to read the 1,484.0 GWh.

A single plant is therefore on the order of a small city connected to a system that already operates with narrow margins and blackout warnings. Two or three plants of this scale on the same salar — which is the horizon the national programme itself contemplates — would be, measured in total energy, on the order of a third of the country's annual generation. It is not a load that a gas-short grid can take, and directing it to the grid would moreover point the demand precisely at the resource Bolivia most lacks.

The consequence is that self-sufficiency ceases to be a constraint of the site and becomes the only architecture available. It is not that the project forgoes a grid it could have used: it is that at this scale dedicated generation is what has to be built, and a connection would not be a substitute for it.

Provenance and caveat. The SIN figures in this subsection come from a third-party energy note of July 2026 which attributes them to the digital yearbook of the AETN (the electricity and nuclear technology regulator) and the CNDC (the national load dispatch committee). They have not been verified against the primary source in this assessment and must be confirmed before being cited to a lender. The percentages relating them to this plant are this document's own arithmetic on the figures of Section 2.

1.2 The solar resource

The Altiplano at 3,700 metres is one of the best solar resources on the planet, and the design rests entirely on that statement. The design direct normal irradiance is 2,700 kWh/m² a year, within a typical band of 2,600 to 2,900 for the region. That value is the most influential parameter in the whole chapter, because the solar thermal field is sized directly on it: Section 3.3 sets out the calculation, in which the aperture is solved so that the annual energy check closes at exactly zero.

The photovoltaic field is sized not on the irradiance itself but on the capacity factor obtained from it with the technology chosen. The design value is **23%** — some 2,015 kWh per kWp installed — and it is the figure consistent with the fixed-tilt structure adopted.

A capacity factor of 27%, some 2,365 kWh/kWp, would require solar trackers, which this design does not use. An independent consultant modelling a 300 MWp project on the same Altiplano takes 2,050 kWh/kWp as a maximum, equivalent to 23.4%, and the design value sits just below that.

The choice is not cosmetic. At the same consumption, 23% rather than 27% is 17% more field, with the capital that carries. The table below runs through the full range on the 283.6 MW field, including the P90 that a lender will reasonably require to see.

CAPACITY FACTOR	TOTAL FIELD	AREA AT 60 MW/KM²	CAPITAL EFFECT	PROVENANCE OF THE VALUE
27% — *rejected*	241.6 MW	4.03 km²	−44 $m	Implies solar trackers, inconsistent with the fixed-tilt structure adopted.

CAPACITY FACTOR	TOTAL FIELD	AREA AT 60 MW/KM²	CAPITAL EFFECT	PROVENANCE OF THE VALUE
23.4%	278.7 MW	4.65 km²	−5 $m	Maximum used by the independent consultant for 300 MWp on the Altiplano.
23% — base case	**283.6 MW**	**4.73 km²**	**—**	Adopted. Some 2,015 kWh/ kWp. No local measurement.
21% — *P90*	310.6 MW	5.18 km²	+28 $m	The model's own P90 estimate. With the current field, it too fits within the allocated land.
19% — *stress*	343.3 MW	5.72 km²	+63 $m	Corresponds to no estimate; included to show the slope.

1.3 The climate that sets the design

Two climatic facts govern the design and push in opposite directions.

The cold

Nights reach −10 °C. For the panels this is an advantage: photovoltaic efficiency improves in the cold, the air is dry, snow is scarce and the morning sun clears the frost. No heating whatsoever is foreseen in the photovoltaic field.

For the batteries it is the opposite, and it is a contract condition, not a recommendation. Lithium iron phosphate cells will not accept charge below 0 °C and degrade outside the 20 to 25 °C band. The system requires active cell heating and liquid thermal management with insulation, not cooling alone. The turnkey price adopted — on the order of 227 $/kWh installed — has been set explicitly to cover that heating, and the procurement specification must say so in those words. Night-time heating moreover introduces a parasitic consumption of its own, small but real, which is not disaggregated in the model.

The wet season

Between January and March there are episodes of three to four consecutive days of dense cloud or snowfall. This is the design event and Section 4 develops it in full. Here it is enough to note the distinction that proved decisive in sizing it, because it is not obvious.

Cloud and snow are not the same problem. With a residual output of 10% of clear sky — dense cloud, clean panels — the 284 MW field delivers some 6.5 MW on average, which covers a little more than half of a 12 MW hold load and reduces the deficit to 5.5 MW. The binding case, then, is not the cloud: it is *snow on the panel*, which takes output to zero. A concrete operational consequence follows: panel clearing becomes a genuine mitigation measure and not a maintenance task, and the cleaning system already costed acquires a second function.

Soiling

The site is a salt surface. Saline dust requires wet cleaning — dry sweeping does not remove it and scratches the glass — at an estimated frequency of 18 to 24 cleanings a year, for both the photovoltaic panels and the mirrors of the solar thermal field. This would be a water supply problem at any desert solar plant. Here it is not, and the reason is developed in Section 6.2.

ALTERNATIVES

Wind: why it does not solve this problem

There is no wind resource assessment for this site, and wind arises naturally as the complement that could cover the design event without burning fuel. The available literature, however, points against it, and this should be said before someone proposes it as the solution.

A peer-reviewed numerical study of the wind potential of the Bolivian Altiplano reports for Uyuni an annual mean on the order of **5.0 m/s at 100 m height**, with a monthly range from 4.3 m/s in April to 5.6 m/s in September. It is below the study's own 7.5 m/s threshold for classifying a site as high-wind. To that must be added altitude: air density at 3,700 m is 69% of that at sea level, and wind power is proportional to density, so that a turbine at 5.0 m/s at this site delivers what it would deliver at some 4.5 m/s at sea level. It is a poor resource, not a marginal one.

And the seasonality runs in exactly the opposite direction to the one required. The same study places the high-wind period between June and October, and the low-wind months between December and May. This project's design event occurs between January and March. The wind is weakest precisely in the window in which it would be needed.

Two qualifications, both honest. The first works in favour: the same study places the daily maximum of wind generation between 20:00 and 22:00, which is when the battery begins to discharge — so that even a poor resource would relieve the night-time peak all year round. The second is a caveat about the statistic: the monthly mean is not the figure that decides this question. What matters is the wind *during* episodes of irradiance below 2 kWh/m² a day, and a front bringing dense cloud or snow may perfectly well bring wind. That datum does not exist for this site and cannot be inferred from a monthly mean.

The conclusion of this assessment is therefore that wind moves from “long-term answer” to a **hypothesis to be discarded or confirmed by measurement**, and that the measurement must answer a specific question — how much wind there is during the sunless episodes — and not the generic one of what the annual mean is. No figure in this document incorporates it.

SECTION 02

2. The demand

The plant's energy demand is made up of three blocks which the model handles in separate tabs and which are consolidated here. The separation has a reason — avoiding double counting, which in a model with cross-links is a real risk — but it leaves the reader without a single figure, and that single figure is what a lender needs.

> **The evaporation basis.** The 2026 technical review of the evaporation design establishes it: 49 tonnes of water evaporated per tonne of carbonate — 3.92 Mm³ a year — in closed four- to five-stage crystallisers driven by solar steam, with recompression that reuses each tonne of steam 3.5 times. The brine is not boiled with electricity: the only electrical load the evaporation carries is the compressor.

2.1 The plant's electrical load

The continuous load of the plant and auxiliaries is 45.0 MW, made up as follows:

COMPONENT	LOAD	CONTENT
Process plant	40.0 MW	Pumping, processing, compression, reagent handling. Excludes the recompression compressor, which is accounted for separately in 2.2. It is a hand-entered figure in the model, with no disaggregation.
Wellfield pumping	2.0 MW	Brine abstraction and the 18 km pipeline to the plant.
Camp	2.0 MW	Accommodation, services, heat tracing.
Solar thermal field parasitics	1.0 MW	Heat transfer fluid pumping and collector tracking.
Total continuous load	**45.0 MW**	**8,760 h/y — 394.2 GWh/y**

> **Two loads are not in the table above, and should be.** The model moves **3.67 Mt/y of salt 38 km** to the disposal cells by elevated electric conveyor. At 0.08–0.15 kWh per tonne-kilometre that is 11–21 GWh/y, some **1.3–2.4 MW continuous**, running all year because the plant produces salt all year. It is either contained within the 40.0 MW hand-entered figure or absent from it; the model does not disaggregate that figure, so it cannot be told which. Separately, the residual magnesium liquor — 1.47 Mm³/y to deep-well injection — carries a pumping load on the order of **0.2 MW**. Neither is large against 49.5 MW, but both are named loads that the table does not name, and the magnesium route is unresolved: see 6.5.

The three auxiliaries were explicitly incorporated in a later revision; before it they were not counted, and the omission was worth 5 MW, that is 12% of the load. It is noted because it illustrates the direction in which the corrections to this model have moved.

2.2 Evaporation of the brine

The pond-free flowsheet concentrates the brine in closed crystallisers instead of in open-air ponds. That turns a process which was free and slow into one which is fast and energy-intensive, and it is the largest load in the project. What changed in August is *what kind* of energy that is.

The evaporation work is done by **steam**, produced in the same solar thermal field that supplies the finishing. Mechanical recompression does not replace that steam: it economises it, reusing each tonne 3.5 times before condensing it. The arithmetic is as follows.

$$3{,}920{,}000\ m^3/y \times 627\ kWh\text{-}t/m^3 \div 3.5 = 702.2\ GWh\text{-}t/y$$

Water to be evaporated × latent heat of water ÷ steam reuse factor

That is **80.2 MW-t continuous** of steam, 127.9 t/h, which Section 2.3 adds to the thermal load and Section 3.3 converts into solar field. The compressor that achieves the reuse has an electrical load of its own, much smaller:

$$3{,}920{,}000\ m^3/y \times 10\ kWh\text{-}e/m^3 = 39.2\ GWh/y$$

Water to be evaporated × specific consumption of the recompression compressor

That is **4.5 MW continuous**. The volume to be evaporated — 49 t of water per tonne of carbonate — is derived from that review: 30 t up to the solubility limit of lithium sulfate and the potassium salts, plus 19 t in the Route C stage. Of the 58 t of water that enter with the brine, some 13 leave as water of crystallisation within the carnallite and the epsomite and are never boiled. Along the way 5 t of salt crystallise per tonne of product: 400 kt a year, discharged continuously and with no quality specification.

TRACEABILITY

Where this figure comes from, and where it does not

A reviewer opening the workbook will find three different sizings of this same circuit, and it should be said which one governs. The first, in the *Water Recovery* tab, arrived at 49.7 MW of photovoltaics on 3.92 Mm³, 30 kWh/m³ and a capacity factor of 27%: a self-contained sizing that is **inactive** and feeds nothing. The second, on 5.38 Mm³ and 35 kWh/m³ entirely electrical, is an all-electric train that the pond-free flowsheet does not use. The third is the one in force: 3.92 Mm³, solar steam and a 10 kWh/m³ compressor, and it is the one that feeds the Dashboard and every figure in this version.

Two labels in the workbook are out of date and should be corrected before they circulate. The cell that adds up the batteries says “plant 720 + MVR 344” against a total of 791.6 MWh: the second addend is today 71.6, not 344. And the sensitivity note on steam reuse states that at 3.0 reuses the field falls to 315 MW-t; the arithmetic runs the other way — less reuse is more steam — and the correct value is 432.5 MW-t. Neither of the two feeds a formula; both feed a third party's reading.

2.3 The thermal load

The heat is taken from a solar thermal field and not from fuel-fired boilers, which is what most lithium plants do. The field covers the finishing — preheating, impurity removal, carbonation and precipitation, washing and drying — and the evaporation steam of Section 2.2. The second is much the larger of the two.

STAGE OF THE CALCULATION	VALUE	BASIS
Gross process heat	50.99 MW-t	Sum of the stages of the flowsheet.
Heat recovery credit	22%	Exchanger network and condensate reuse.
Net finishing load	39.77 MW-t	Gross less the recovery credit.

STAGE OF THE CALCULATION	VALUE	BASIS
Campaign peak factor	1.00 ×	Set at 1.0: with no ponds the plant is fed with brine abstracted continuously and runs flat all year.
Brine evaporation steam	80.16 MW-t	Section 2.2.
Total net thermal load	**119.93 MW-t**	**Sizing basis for the solar thermal field.**
Annual thermal energy	**1,050,622 MWh-t**	**Total load × 8,760 h of operation, the pond-free route running flat all year.**

Setting the campaign factor at 1.0 saved on the order of USD 55 million of solar field, and it is defensible: with no ponds, the plant is fed with brine abstracted continuously and has no seasonality of feed. Should the process engineer later demonstrate a genuinely seasonal thermal profile, the factor rises again and the field with it.

The proportion is worth seeing clearly, because it sets the hierarchy of the document. Of the 119.9 MW-t, the finishing is 33%. The steam that evaporates the brine is the remaining 67%. The solar thermal field is the largest single piece of the supply — in power, in area and in capital.

2.4 The consolidated balance

Adding the blocks, and including the cloud reserve built into the plant photovoltaic field, the project's annual balance is as follows.

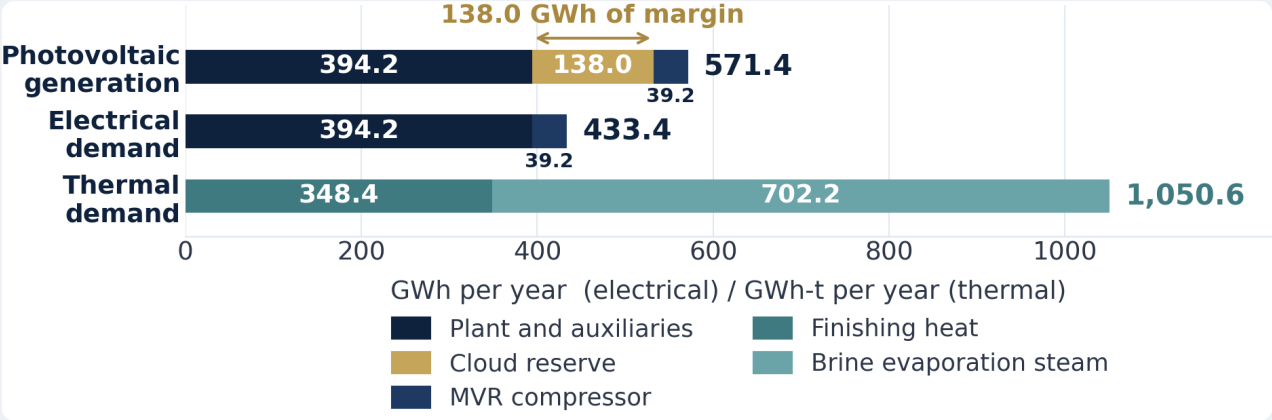


Figure 1 — Consolidated annual energy balance on the current basis. Photovoltaic generation exceeds electrical demand by 138.0 GWh: this is the reserve intended to recharge the batteries on marginal days, not energy available for any other use. The thermal bar and the two electrical bars are in different units and must not be read as equivalent in quality: a thermal MWh of solar steam does not substitute for an electrical MWh.

ITEM	POWER	ANNUAL ENERGY	PER TONNE
Plant and auxiliaries	45.0 MW	394.2 GWh	4,928 kWh/t
Mechanical vapour recompression compressor	4.5 MW	39.2 GWh	490 kWh/t
Total electrical demand	**49.5 MW**	**433.4 GWh**	**5,418 kWh/t**
Finishing heat	39.8 MW-t	348.4 GWh-t	4,355 kWh-t/t

ITEM	POWER	ANNUAL ENERGY	PER TONNE
Brine evaporation steam	80.2 MW-t	702.2 GWh-t	8,778 kWh-t/t
Total thermal demand	**119.9 MW-t**	**1,050.6 GWh-t**	**13,133 kWh-t/t**
Total energy demand	**169.4 MW**	**1,484.0 GWh**	**18,550 kWh/t**

INDEPENDENT CROSS-CHECK

An external estimate arrives at the same order

A third-party energy note on Potosí, prepared in July 2026 from public life-cycle studies, places the energy intensity of a brine lithium carbonate plant at between 5 and 15 MWh per tonne, with a central value of 10. **This project produces 18.55 MWh per tonne, 24% above the upper end of that range**, and that divergence is declared here rather than left implicit.

The explanation is arithmetic and not mysterious. Evaporating a cubic metre of water with solar steam reused 3.5 times costs some 179 thermal kWh; doing it with fully electric recompression costs 35 electrical kWh. That is five times as many units of energy for the same physical work, because recompression harnesses mechanical work and steam does not. The first path is the one the chemistry of this brine admits; the second is the one a pond mass balance made appear available.

Two consequences, and they should be held together. The first is that the comparison with the literature loses value: a range built on direct extraction plants, mostly electrical, is not the right benchmark for a thermal route. The second is that the additional energy is of the cheap class: 63 dollars per thermal MWh against 139 per electrical MWh.

Eighteen thousand five hundred kilowatt-hours per tonne of carbonate is a high figure even for a pond-free route, and the reason is the same as ever taken one step further: eliminating the evaporation ponds substitutes free, slow solar energy with bought, fast energy, and here that bought energy is steam and not electricity. Section 5 shows where the balance lands.

SECTION 03

3. The architecture of the supply

The supply is made up of four elements that perform distinct functions and which should not be confused with one another: the photovoltaic field produces the electricity, the batteries carry it into the night, the solar thermal field produces *all* the heat — the finishing heat and the steam that evaporates the brine — and carries it into the night on its own account, and the diesel gensets produce no useful energy in any normal operating scenario. Of the four, the solar thermal field is much the largest.

3.1 The photovoltaic field

BLOCK	POWER	ANNUAL ENERGY	FUNCTION
Plant field — base	195.7 MW	394.2 GWh	Covers the continuous load of the plant and auxiliaries.
Oversizing for cloud	68.5 MW	138.0 GWh	35% oversize. Allows the battery to be recharged on marginal days, not merely the instantaneous load to be covered.
Recompression compressor block	19.5 MW	39.2 GWh	Covers the compressor of Section 2.2. Sized to the annual energy; no margin of its own was added.
Total field	**283.6 MW**	**571.4 GWh**	**A single field. The three are sizing blocks, not three installations.**

Fixed-tilt structure, not trackers

The decision to mount on fixed-tilt structure instead of solar trackers is explicit and is taken against the performance argument. Trackers would give a capacity factor of about 27% and would reduce the field by some 42 MW. They were discarded for the reason that applies at a site like this one: they are moving parts in an environment of salt, dust, wind and −10 °C, four hours from the nearest town, at a plant that cannot buy energy when its own field is out of service. Fixed-tilt structure has nothing that can seize.

The cost of that decision — 17% more field and some USD 44 million — is incorporated in every figure in this document. It is a good example of a trade-off between capital cost and availability that at a grid-connected plant would have been resolved the other way.

Local precedent

The Uyuni Altiplano already hosts a **60 MW** solar plant operated by ENDE Guaracachi (a subsidiary of the state electricity company). This project's field is on the order of five times that installation, so this is not a technology without precedent at the site but a step-change in scale on a precedent that exists. It is the first place to look for real measured performance, construction costs at this altitude and cleaning and soiling experience, and it should be consulted before the contractor quotation of Section 8.3.

A contrast of criteria should also be noted. The same third-party note that documents that plant estimates that covering 90 MW continuous would require on the order of 330 MW of photovoltaics, which implies a capacity factor of 27.3%. It is exactly the value this model moved away from as optimistic, and applied to this project's 49.5 MW continuous it would give a field of 181 MW instead of 284. **The sizing in this**

document is therefore 56% more conservative than the external estimate, and the difference is the capacity factor of Section 1.2 and the cloud reserve of 3.2.

The unit cost

The model uses 0.73 $/W of direct cost, which with the loadings for engineering, contingency and the remoteness and altitude premium reaches on the order of 1.10 $/W installed. It is a high figure against international prices and it is deliberate: everything is imported, there is no local supply chain, and the works are at 3,700 metres. The figure must be confirmed with a contractor quotation.

One field, three calculation blocks

A misunderstanding that the structure of the model may induce is worth avoiding. The three blocks in the table above are calculated in two different tabs — the plant block and its reserve in one, the compressor block in another — because they serve different loads and are costed against different line items. They are neither three fields nor two fields: **they are a single 283.6 MW field**. The model itself consolidates them under the heading “total solar field, all the photovoltaics, in one place”, reads the batteries as a single fleet of 791.6 MWh, scales the cleaning cost on the combined capacity, and the master plan reserves a single fenced enclosure. This document does the same: it treats them as one field for area, cleaning, reserve and firm supply, and disaggregated only where the financial model disaggregates them. What remains open is not whether there are one or two fields, but how it is contracted: a single 283.6 MW tender obtains economies of scale in civil works, medium-voltage architecture, control room and operating contract that two separate packages do not.

3.2 Battery storage

BLOCK	CAPACITY	AUTONOMY	NOTE
Plant battery	720.0 MWh	16 h	On 45 MW of load. Raised from 270 MWh / 6 h in a later revision.
Recompression compressor battery	71.6 MWh	16 h	On 4.5 MW of load.
Total storage	**791.6 MWh**	**16 h**	**Lithium iron phosphate, with active cell heating.**

Sixteen hours of autonomy is not a conventional figure. A grid-connected system usually carries two to four. Here it covers the whole night with margin, and that margin exists because there is no other source to turn to at four in the morning. The increase from six hours was one of the structural corrections to the design and by itself accounts for some USD 95 million of the installed capital.

The sixteen hours are sixteen hours of the operating load, not of the hold load. In normal operation that is the design: the battery covers the night. In the design event the plant is not at 49.5 MW but at the 12 MW hold load, and the same fleet then lasts 44.5 hours — which is what the firm-supply case is built on, and is developed in Section 4.3.

Autonomy and recharge are two different things

Two questions that the “16 hours” figure tends to merge into one should be separated. The first is how long the battery lasts. The second is how long the field takes to fill it again. The first is answered by the storage; the second only by the oversizing, and that is why the oversizing exists.

HOW LONG THE BATTERY LASTS	ENERGY	DURATION	NOTE
At full operating load — 49.5 MW	791.6 MWh	16.0 h	It is the design parameter: it covers the night, not an event.
In safe hold — 12 MW, entering at 75% state of charge	534.3 MWh	44.5 h — 1.9 days	The figure the model uses to size the backup.
In safe hold — 12 MW, entering at 100%	**712.4 MWh**	**59.4 h — 2.5 days**	**With forecast pre-charging. It is here that "two and a half days of battery" lives.**
To sustain full load for 3 days would require	3,562 MWh	72 h	4.5 times what is installed. It is not the design and does not claim to be.

The two and a half days do exist, then, and they are real — but they are two and a half days *of a stopped and protected plant*, with the battery deliberately filled beforehand, and not three days of production. That is exactly the logic of Section 4, and it is what makes the diesel backup cost USD 19 million and not USD 107 million. The 59.4 hours are reached only with forecast pre-charging committed in writing.

The second question is the one the oversizing answers, and its arithmetic is as follows. The 283.6 MW field generates on average 1,565 MWh a day; the plant consumes 1,187. The surplus available for recharging is **378 MWh a day**, so that replenishing the 712 MWh of usable energy from an empty battery requires **a little under two days of average sun**. Put another way, and this is the operating figure that matters: **the field covers the load on a day delivering 75.9% of mean irradiance**. Any day below that threshold does not recharge: it discharges.

> **Recovery, not endurance.** The oversize is not what carries the plant through the worst event — the battery and the backup do that. It is what refills the battery afterwards, and it is why Section 4.7 flags consecutive events as an unmodelled case: after a four-day event the system needs almost two days of normal sun to regain the autonomy attributed to it.

What it costs to buy more firm supply

Should it be concluded that 16 hours at full load are not enough, it is worth knowing what each dollar buys. The four available routes are not comparable with one another in what they deliver, and only one of them recurs over time.

ROUTE	CAPITAL	LIFE-CYCLE COST	WHAT IT ACTUALLY DELIVERS
Safe hold gensets — *already adopted*	18.7 $m	18.7 $m	Covers the tail of a four-day event at hold load, indefinitely. It is by far the cheapest firm supply per hour, and it is already bought.
Battery from 16 h to 20 h — +198 MWh	**41.6 $m**	**97.2 $m**	**Takes autonomy in hold from 44.5 to 55.7 hours at 75% state of charge, and the annual hours of the gensets from 205.9 to 161, without depending on a procedure.**
Battery from 16 h to 24 h — +396 MWh	83.1 $m	**194.4 $m**	Eight more hours at full load. The life-cycle cost includes two cell replacements: it is the only component that is bought again.

ROUTE	CAPITAL	LIFE-CYCLE COST	WHAT IT ACTUALLY DELIVERS
Battery from 16 h to 48 h — +1,583 MWh	332.5 $m	**777.8 $m**	Two days at full load. On the order of a tenth of the total project cost, for a case that safe hold already resolves for USD 18.7 million.
Full continuity with diesel — 117 MW	107 $m	107 $m + 200 $/t	Costed and discarded in August 2026. It protects production, which is what the project can afford to lose.
Wind	—	—	Not costed. The local resource is poor and its high season is the opposite of the design event — see 1.3 and 3.5.

Three readings follow from the table. The first is that **storage remains the most expensive way to buy endurance**, and doubly so because the cells are replaced twice in the life of the project while the panels and the gensets are not. The second is that the two cheapest measures do not appear in the table because they cost no capital: forecast pre-charging adds some fifteen hours of autonomy in hold, and mechanical panel clearing attacks the binding case directly, which is snow on the glass and not cloud. The third is that there is a concrete and bounded reason to buy battery — four hours, USD 42 million — which is not endurance in itself but the emergency-plant classification of the fossil backup. Section 5.5 values it against the procedural alternative, which is free.

> **The prior question.** None of the above makes sense without first settling a question of intent, and it is not a technical one: if the objective is still to *protect the plant*, then 16 hours at full load is not a defect — it is the design, and the relevant autonomy is the 1.9 to 2.5 days in hold. If the objective has become *sustaining production* through a multi-day event, then this is not a matter of adjusting this system but of a different project, and it has already been costed: USD 107 million and 200 dollars per tonne. That should be decided before anything is sized.

3.3 The solar thermal field

The heat is produced with solar concentration and molten-salt storage. It is a less usual decision than the photovoltaics and it merits explanation: most lithium plants burn diesel or gas for this function, and at this site that would have meant **123.6 million litres of diesel a year**, hauled along a route that floods a third of the year.

The field covers both the finishing — 39.8 MW-t — and the steam that evaporates the brine — 80.2 MW-t. Two points of method should be stated before the table. The first: the solar multiple is not a fixed value but the greater of 2.8 and whatever the *annual energy balance* requires, so that the field is sized against the year and not against the peak alone. The second: the field runs 8,760 hours, flat, because the pond-free route has no seasonality of feed.

PARAMETER	VALUE	NOTE
Finishing heat	39.77 MW-t	Section 2.3.
Brine evaporation steam	80.16 MW-t	Section 2.2.
Total net design load	**119.93 MW-t**	**Sizing basis.**
Solar multiple	3.24 ×	The greater of 2.8 and what the annual energy balance requires.
Thermal peak of the field	**389.12 MW-t**	**Load × solar multiple.**
Design direct normal irradiance	2,700 kWh/m²/y	Literature value, not measured at the site. It sets the aperture directly.

PARAMETER	VALUE	NOTE
Net optical-thermal efficiency	55%	Collector optics net of thermal losses.
Collector aperture	**0.7075 km²**	**707,490 m². Some 1.14 times the solar thermal field of the Zabuye lithium plant, in Tibet.**
Annual thermal energy of the field	1,050,622 MWh-t	Equal to the annual demand: the aperture is solved from this balance. **Annual energy margin: zero.**
Molten-salt storage	15 h — 1,799.0 MWh-t	On the total load. It covers the night and, as Section 4.3 shows, the entire design event at hold load.
Fenced area allocated	1 km²	By master plan.

Scale-up is an open point: this project is on the order of four times the tonnage of Zabuye, and that the field scales linearly — or better — must be confirmed with the solar field designer.

> **The annual energy check closes at zero, on an irradiance not yet measured.** This is not the same as the field being undersized in capacity: the solar multiple of 3.24 is precisely the capacity margin, and it is what charges the fifteen hours of molten-salt storage on this same field. What the workbook declares is the annual balance: the thermal energy the field delivers in a year — 1,050,622 MWh-t — is exactly equal to what the plant demands, because the solar multiple was raised until that equality held. That result rests entirely on a direct normal irradiance of 2,700 kWh/m² a year which is from the literature. At 2,600 — within the regional band — the required aperture rises to 734,700 m² and the peak of the field to 404 MW-t; at 2,900 it falls to 658,700 m² and 362 MW-t. It is a ±4% sensitivity on the capital of the largest component of the system, and it is closed with a measurement station, not with a discussion. *Traceability note:* a cell note in the workbook states that at 2,600 kWh/m² the aperture would be 544,000 m²; that figure does not follow from the formula in the tab itself and has not been used here.

The levelised cost of the resulting heat is examined in Section 5.3, where a fuel price that the model keeps out of date in favour of diesel is also addressed.

3.4 The firm-supply reserve

This is the element a credit insurer will examine most closely, and it is the one with the least obvious logic, so the principle is stated first and the sizing afterwards.

DESIGN PRINCIPLE

The right question is not how to keep producing

Faced with a multi-day event without sun there are two possible questions. The first is “how do we maintain production”, and its answer costs USD 107 million and 117 MW of fossil generation. The second is “how do we guarantee that we never suffer an unplanned hard shutdown”, and its answer costs USD 18.7 million and 21.2 MW.

The risk the sponsor described is not losing tonnage: it is plugging a column, freezing a crystalliser, and spending weeks clearing it. That is a risk *to the plant*, not to production, and it is eliminated by guaranteeing an orderly drain-down and shutdown and keeping the lines above the crystallisation temperature. The production of those days can be lost. The asset cannot.

PARAMETER	VALUE	NOTE
Load to be sustained in hold	12 MW	Brine circulation, agitators, heat tracing, instrument air, control, camp, freeze protection.
Load shed	76%	From 49.5 MW in full operation to 12 MW in hold. It was 82% when the full load was 66.5 MW.
Maximum continuous load factor	85%	Of the generating set.
Altitude derating at 3,700 m	72%	Of the ISO rating. Requires a vendor-certified curve at −10 °C.
N+1 redundancy	8%	One spare block.
Installed nominal capacity	**21.2 MW ISO**	**Seven containerised 3.5 MW units at medium voltage.**
Annual running	**205.9 h**	**Emergency running only; the gensets carry no production — Sections 4.4 and 5.5.**
Tank farm	772 m³	With a 25% margin. Covers the whole wet season.

Seven containers and a tank of eight hundred cubic metres. It is an installation, not a power station, and that difference of scale is what keeps the matter out of the environmental conversation. Section 4.5 sets out the two measures that reduce the hours it runs.

3.5 What was not chosen

Four alternatives were examined. Two were discarded, one was eliminated as non-existent, and one remains open.

OPTION	STATUS	REASON
Grid at YLB cost	Eliminated	There is no reliable grid at the site. The mode was in the model and was described as fiction for this site. It is locked out.
Full production continuity	Discarded	117 MW ISO — 34 gensets of 3.5 MW. On the 49.5 MW operating load it would be some 87 MW ISO; the 117 MW costing is retained because it is the one against which the decision was taken, and it is the conservative of the two. Capital 107 $m, operation 200 $/t, 6.15 million litres a year, 7,700 m³ of storage. It costs 0.076 of minimum debt service coverage and 70 basis points of return. It was not discarded on economics — the difference against doing nothing was within the noise of a ±40% estimate — but because it buys what is not needed: it protects production, when the real risk is to the plant. It also creates new risks: a 117 MW fossil plant with a real duty cycle is a different proposition environmentally, and 7,700 m³ of bulk fuel within the Salar de Uyuni basin is an environmental schedule risk.
Nuclear microreactors	Discarded as a design option	It is not financeable for this project. There is no commercial licence, no commercially deployed reactor anywhere, and no published price per kilowatt. Units of this type are offered at around 5 MW: the 45 MW plant load is nine units, and the 49.5 MW of the whole is ten, of a design that commercially does not exist, and nuclear technology would then have to be licensed in Bolivia, before a regulator with no precedent, within the schedule on which the entire financing depends. It remains alive as a **repowering option** in the year-20 cell replacement decision, not as a decision for today.

OPTION	STATUS	REASON
Wind	Open — unfavourable in the literature	An annual mean of some 5.0 m/s at 100 m for Uyuni, below the 7.5 m/s high-wind threshold, and with air density at 3,700 m at 69% of that at sea level. Worse still, the high-wind period is June to October and the design event occurs from January to March: the resource is weakest when it would be needed. In its favour is its daily maximum from 20:00 to 22:00, which relieves the night-time peak. There is no local measurement, and the question it must answer is how much wind there is *during* the sunless episodes, not what the annual mean is. Sections 1.3 and 8.2.

SECTION 04

4. The design event

This section answers the one question a self-sufficient system cannot evade: what happens when the resource disappears for several days in a row. Everything else in this document — the size of the field, the size of the battery, the cost — is a consequence of how it is answered here.

4.1 Method

The method is an energy balance over a discrete event, not an hourly simulation of the year. A design event is defined with a duration, an initial condition and a residual output; the stored energy available at the start is calculated; the sustained load is deducted hour by hour; and the moment at which the storage is exhausted is identified. What remains between that moment and the end of the event is the deficit, and it is the only thing the fossil backup has to cover.

It is a deliberately simple method and its virtue is that it can be followed by hand. Its limitation is set out in Section 4.7.

4.2 The event

PARAMETER	VALUE	BASIS
Duration	4 days	The sponsor states episodes of 3 to 4 consecutive days of dense cloud or snowfall.
Events per wet season	4	Between January and March. It determines the fuel and the size of the tank, not the installed capacity.
Battery state of charge at the start	75%	The event is not always entered with a full battery.
Usable depth of discharge	90%	Lithium iron phosphate cells cannot be taken to zero.
Residual photovoltaic output	**0%**	**Snow on the panels. This is the binding case — see below.**

Why zero and not ten per cent. With a residual of 10% — dense cloud with clean panels — the 283.6 MW field delivers some 6.5 MW on average, which covers a little over half of the 12 MW hold load and leaves a deficit of 5.5 MW. Cloud is the benign case and snow the binding one: the residual under cloud covers half the hold load, the residual under snow covers none of it. Sizing against zero residual is sizing against snowfall, which is correct, and *mechanical panel clearing* is a first-order mitigation measure not formalised today as a procedure.

4.3 The autonomy already bought

This is the result that collapses the problem, and it is the reason the backup costs USD 19 million and not USD 107 million.

ITEM	VALUE	NOTE
Installed batteries	791.6 MWh	Plant 720 + recompression compressor 71.6. *The label in the workbook still says "plant 720 + MVR 344" — see 2.2.*
Available at the start of the event	534.3 MWh	After applying the initial state of charge and the depth of discharge.
Autonomy on batteries alone, at hold load	**44.5 h**	**Less than two days.**
Installed molten salt	1,799.0 MWh-t	15 h at full load; much more at hold load.
Available at the start of the event	1,349.3 MWh-t	
Autonomy on salt alone, at hold load	**112.4 h**	**It covers the full 96 h of the event with 189 MWh-t to spare: there is no thermal deficit.**

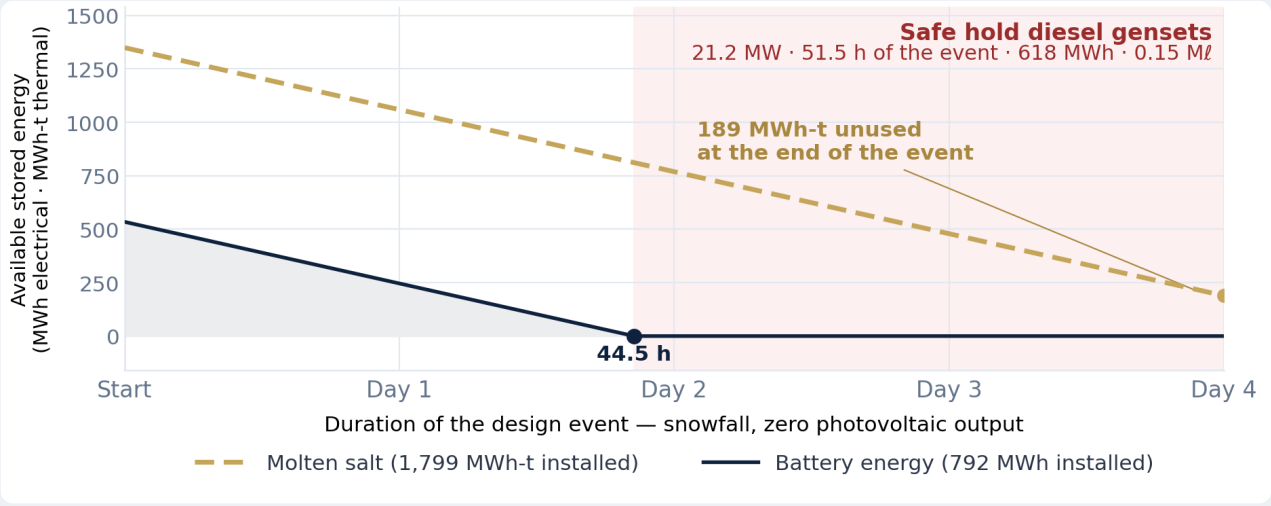


Figure 2 — Depletion of storage during the four-day design event, with zero photovoltaic output and the plant in safe hold at 12 MW electrical and 12 MW-t thermal, on the current basis. The batteries cover the first 44.5 hours on their own; the molten salt, sized on the total thermal load, covers the entire thermal event and reaches the end with 189 MWh-t unused. The diesel gensets cover the shaded band — 51.5 of the 96 hours, somewhat more than half the event.

4.4 The deficit and what it costs to cover it

ITEM	VALUE	NOTE
Electrical demand during the event	1,152 MWh	12 MW × 96 h.
Electrical deficit — covered by the diesel	617.7 MWh	Per event.
Thermal deficit — covered by the boiler	0 MWh-t	Per event. The molten salt covers the entire thermal event — see 4.3.
Fuel per event	0.154 million ℓ	Electrical only: the molten salt carries the thermal side.
Fuel per year	**0.618 million ℓ**	**Against 6.15 million under full production continuity.**

ITEM	VALUE	NOTE
Running hours per year	**205.9 h**	**The figure cited to the credit insurer.**
Emissions of the backup	1,655 t CO_2/y	0.5% of what the solar thermal field saves.

4.5 The two measures that cost no capital

Everything above has been calculated assuming that neither of the two measures in this section exists. The gensets were sized on that basis, and for procurement purposes it is right that they should be: emergency equipment is not contracted against procedures that have not yet been written. But it is worth quantifying what those measures would deliver, because it appears in no figure in the model and because both are free.

Forecast pre-charging

The design event assumes the battery is at 75% state of charge when it begins. It is a realistic assumption: on an ordinary day nothing obliges the system to fill the pack, and holding lithium iron phosphate cells permanently at 100% accelerates their calendar ageing, so that a setpoint fixed at 100% would be counterproductive. With a reliable 48-hour forecast, however, it is known that the front is coming and the pack is filled deliberately.

STATE OF CHARGE AT THE START OF THE EVENT	USABLE ENERGY	AUTONOMY AT 12 MW
791.6 × 0.75 × 0.90 — *model assumption*	534.3 MWh	44.5 h
791.6 × 1.00 × 0.90 — *with pre-charging*	712.4 MWh	59.4 h
Gain	**+178.1 MWh**	**+14.8 h**

Fifteen hours of autonomy in exchange for a decision. No storage is bought: what is bought is *scheduling* on storage already owned. For it to be real, little is needed and it is needed in writing: someone responsible for watching the forecast, a defined threshold that triggers the pre-charging, and an authorisation to execute it. None of that is committed today, and that is why the model carries 75% and not 100%. **This measure is not an optional improvement**: it takes the annual hours of the backup from 205.9 to 146.5 on its own, at no capital cost.

Panel clearing

Its logic is the one already set out in Section 4.2: the residual output was set at zero *not* because there is no light during the event, but because there is snow on the glass. The field delivers 65.2 MW of average output; under dense cloud with clean panels it would deliver 10% of that figure, some 6.5 MW, against a hold load of 12 MW. The deficit the diesel exists to cover is not caused by the cloud: it is caused by the deposited snow.

It follows that clearing panels during an event is not maintenance. It is a generating asset, and its effect is proportional to the area cleared.

STATE OF THE FIELD	CONTRIBUTION	DEFICIT	AUTONOMY AT 75%	WITH PRE-CHARGING
Snow on the glass — *design case*	0 MW	12.0 MW	44.5 h	59.4 h
25% of the field cleared	1.6 MW	10.4 MW	51.5 h	68.7 h

STATE OF THE FIELD	CONTRIBUTION	DEFICIT	AUTONOMY AT 75%	WITH PRE-CHARGING
50% of the field cleared	3.3 MW	8.7 MW	61.1 h	81.5 h
Whole field cleared, still cloudy	**6.5 MW**	**5.5 MW**	**97.5 h**	**130.1 h**

The design event lasts 96 hours. **Pre-charging and the whole field cleared reach 130 hours** — enough to close the event without diesel. With half the field, 81.5 hours are reached and the gensets start in any case, though for 15 hours and not for 51. Clearing the 4.7 km² of the whole field is not the same as clearing half of it, and that is what the mitigation asks for.

> **Four caveats, and they are substantive.** Wet washing at −10 °C is exactly the wrong manoeuvre: it would freeze the array. Clearing snow is a mechanical operation, distinct from the saline dust cleaning for which the system of Section 6.2 was costed, and it is not costed. The fixed-tilt structure is a disadvantage here: a tracker can fold to a vertical position to shed the snow and a fixed row cannot, which is a real cost of that decision that is nowhere valued. Five square kilometres of glass are not cleared in a day, so the realistic approach is to clear a priority block sized to the hold load, with the sequence planned in advance and not improvised during the event — and the table above shows that this priority block would have to be practically the whole field in order to close the event without diesel. And all of the above rests on the 10% residual output, which is an assumption and not a measurement — closing it forms part of the programme of Section 8.2.

The honest reading is uncomfortable in both directions. On the one hand, the USD 18.7 million in gensets could be insuring a case that two written procedures would largely dissolve. On the other, one does not give up emergency backup on the strength of two procedures that do not yet exist and of a residual output that has not yet been measured. What does follow clearly is the order: committing to the procedures and measuring the residual costs practically nothing and should be done now; resizing the backup downwards only makes sense afterwards, and with the measurement in hand. The order is urgent for one reason: forecast pre-charging is what decides whether the backup is presented as an emergency plant or not.

4.6 Sensitivity to the hold load

The whole section rests on a figure — the 12 MW hold load — which is an engineering estimate and not the result of a process balance. It is therefore worth showing what happens if the process engineer returns another number.

> **Provenance of this table.** The *Firm Supply (Diesel)* tab of the model contains its own sensitivity table, but its autonomy column is calculated on a 1,063.9 MWh battery fleet and still shows 59.8 h at 12 MW. The nominal capacity and capital columns remain valid, because they depend only on the hold load. Those for autonomy, operation and running hours are recalculated here on the 791.6 MWh actually installed, with the same method as the workbook. The annual hours column does not exist in the model and is this document's arithmetic.

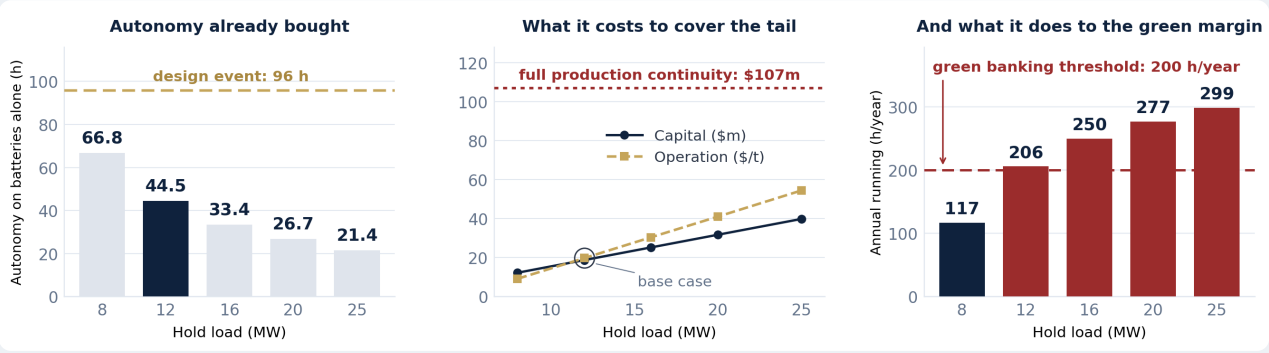


Figure 3 — Sensitivity to the hold load. On the left, the autonomy delivered by the batteries already bought; on the right, the capital and the operating cost of the backup. The case remains cheap and remains advantageous across the whole range examined, which is why the *concept* can be committed to now and the *size* can be signed off later.

HOLD LOAD	BATTERY AUTONOMY	NOMINAL CAPACITY	CAPITAL	OPERATION	HOURS PER YEAR	CONFIGURATION
8 MW	66.8 h	14.1 MW	12.2 $m	9.1 $/t	117 h	5 gensets, 292 m³ of tanks
12 MW	**44.5 h**	**21.2 MW**	**18.7 $m**	**19.7 $/t**	**206 h**	**7 gensets, 772 m³ — base case**
16 MW	33.4 h	28.2 MW	25.2 $m	30.4 $/t	250 h	9 gensets, 1,252 m³
20 MW	26.7 h	35.3 MW	31.7 $m	41.0 $/t	277 h	11 gensets, 1,732 m³
25 MW	21.4 h	44.1 MW	39.8 $m	54.4 $/t	299 h	13 gensets, 2,332 m³

Even at the pessimistic extreme of 25 MW, the backup costs USD 39.8 million — little more than a third of full continuity — and the battery autonomy is still almost a full day. **The conclusion holds across the range.** The safe hold solution remains robust against the parameter that is worst known, and that is a property full continuity does not have, because its cost scales directly with the load in full operation.

4.7 What these results establish and what they do not

This is the point in the document where it is best to be most rigorous, because Section 4 produces precise-looking figures from assumptions that have not been measured.

- **The duration of the event is a statement, not a record.** “Three to four consecutive days” comes from the sponsor. Neither the satellite record nor a typical meteorological year for the site has been consulted. The actual frequency of episodes with irradiance below 2 kWh/m² a day between January and March, and how often the snow actually settles on the panels rather than merely clouding over, are data that exist and that have not been obtained. It is the first measurement in Section 8.2.
- **The hold load is an engineering estimate.** Twelve megawatts is an informed judgement about what must keep running. It does not come from a process balance. Section 4.6 shows that the case withstands the range, which is not the same as saying that the figure is correct.
- **Consecutive events are not modelled.** The calculation assumes four events separated by sufficient recharging windows, and that sufficiency is not free: replenishing the battery from empty requires almost two days of average sun (Section 3.2). Two four-day events separated by two poor days have not been examined, and it is a physically plausible case in the wet season. The fuel tank covers it; the 75% initial state of charge might not.
- **Capacity loss between replacements is not modelled.** Cells lose usable energy from day one. Sustaining 16 hours of autonomy through to year fifteen requires either oversizing on day one or accepting thinner night-time cover at the end of each cycle. The photovoltaic oversizing helps with recharging, but it does not restore lost capacity. It is moreover degradation that decides whether the

annual hours of the backup stay under control over time: a fleet delivering 10% less usable energy in year fourteen pushes the deficit of the event — and with it the hours — in the wrong direction.

- **The altitude derating is not certified.** The 72% of ISO rating at 3,700 m is a handbook value. A vendor-certified curve at 3,700 m and −10 °C is required, plus confirmation of black start from a discharged battery, which is precisely the condition in which these gensets have to start.
- **What is established, and is insensitive to the range:** that the storage bought for normal operation covers half of the electrical event and the whole of the thermal event on its own; that the fossil backup sized to the tail is consequently one sixth of the cost of one sized to production; and that the safe hold solution beats doing nothing both in debt service coverage and in return, across the whole range of hold load examined. These three statements do not depend on the event lasting four days or on the hold load being twelve megawatts.

SECTION 05

5. Costs

All the figures in this section are Class 5, with a declared ±40% band. The factors of greatest swing, in order, are: the unit cost of the solar thermal field per square metre of aperture — the largest item in the system — the direct normal irradiance that sets that aperture, the specific consumption of the recompression compressor, the unit cost of the panel installed at 3,700 metres, and that of the battery with cell heating.

5.1 The capital

COMPONENT	$ MILLION	CONTENT
Plant photovoltaic field	142.8	195.7 MW at 0.73 $/W of direct cost.
Plant battery	108.0	720 MWh at 0.15 $m/MWh of direct cost.
Inverters, balance of plant and connection	20.1	8% on panel and battery.
Direct subtotal	*270.9*	
Engineering and contractor	32.5	12% on direct.
Contingency	48.8	18% on direct.
Remoteness and altitude premium	27.1	10% on direct.
Base plant system	**379.2**	
Photovoltaic oversizing (35%)	50.0	68.5 MW additional.
Safe hold gensets	13.3	21.2 MW ISO plus tank farm, direct cost.
Loadings on the reserve	25.3	Engineering, contingency and altitude on the above.
Firm-supply reserve	**88.7**	
Plant electrical system, with reserve	**467.9**	**Of which 101.7 is built in year 1 for the pilot plant.**
Solar thermal field	**586.7**	**The largest item in the system. Net of the fossil boiler island not built — 24.0 — the incremental contribution is 562.7, and that is what the Dashboard reads. Of the 586.7, some 21.8 is built in year 1 for the pilot.**
Photovoltaics and battery of the recompression compressor	46.4	21.4 of panels and 15.8 of batteries, plus 25% contingency. Accounted for within the evaporation system.
Cleaning systems	5.6	4.8 for the photovoltaic field, 0.8 for the mirrors. Scales with the installed capacity.

COMPONENT	$ MILLION	CONTENT
COMPLETE ENERGY SYSTEM	**1,106.7**	**32% of the total project cost of 3,465.4 $m.**

The overall reading is that the solar thermal field, not the photovoltaics, is where the money is. The electrical system — photovoltaics, batteries, firm-supply reserve and the compressor block — is USD 514 million; the solar thermal field alone is USD 587 million. The evaporator proper — USD 40.3 million, which does not appear in this table because it is process equipment and not supply equipment — completes the evaporation system that the Dashboard accounts for at 86.7.

> **The largest component is the least verified.** The USD 586.7 million of the solar thermal field is built on 250 $/m² of collector field, 35 $/kWh-t of salt storage and an aperture that derives from a literature irradiance. None of the three comes from a quotation. A ±40% deviation on it is worth USD 235 million, 7% of the total project cost. Validation with the field designer — point 4 of Section 8.3 — is the largest open piece in the chapter.

A note on the distribution over time, because it affects how the cash flow reads. A portion of the system — USD 101.7 million of the electrical and USD 21.8 million of the thermal — is built in year 1 of the project to supply the 10,000 t/y pilot plant, which is commissioned in year 1, tested in year 2 and commercial from year 3. It is not additional capital: it moves from the plant construction window into the pilot phase, and the total project cost does not change. What must be done from day one is to lay out the site, the high-voltage architecture and the civil works for the complete field, not for the pilot portion.

5.2 The operation

COMPONENT	$/T	$M/Y	NOTE
Operation and maintenance of the plant photovoltaic and battery system	90.0	7.2	It replaces the electricity bill that does not exist.
Additional for the firm-supply reserve	27.7	2.2	8 for the photovoltaic oversizing and 19.7 of direct operation of the backup, including fuel and monthly testing.
Cleaning of panels and mirrors	12.1	1.0	Scales with the installed capacity of the field.
Evaporation system — evaporator, panels and batteries of the compressor	27.4	2.2	3.5% of the capital of panels and evaporator, 2% of that of batteries. Of that total, some 13.3 $/t corresponds to the energy supply part.
NAMED ENERGY LINE ITEMS	**157.2**	**12.6**	**3.2% of the operating cost of 4,867 $/t.**

> **On double counting.** The model automatically charges, on the total plant capital, 0.8% a year of insurance and 1.2% of maintenance materials, plus a 10% operating contingency. The energy system drags those items along by virtue of its size — on the order of 5.1 $/t for the diesel backup alone — and they are already incorporated in the project's total operating cost. They have not been added again in the table above. Anyone reconstructing these figures must bear in mind that the cost shown here is the *direct* cost and that the *total attributable* cost is higher.

5.3 The comparison with diesel

The realistic alternative at this site is neither the grid nor gas: it is diesel generation, which is what remote mining operations on the Altiplano do. That is the appropriate comparison.

The right-hand column is a counterfactual. It describes a plant that is not built and a fuel that is not bought. The project acquires diesel for one reason only and in one quantity only: 0.62 million litres a year of emergency fuel, burned in some 206 hours of safe hold. Everything else in that column is what it would have cost to make this same energy another way, and it is shown to size the decision, not to be added to anything.

ITEM	PROJECT SYSTEM	DIESEL EQUIVALENT	NOTE
Levelised cost of electricity	139 $/MWh	438 $/MWh	The diesel figure is *fuel only*, with no capital or maintenance, and at the delivered price for small volumes.
Levelised cost of heat	63 $/MWh-t	206 $/MWh-t	At the delivered price in force. The solar thermal field tab shows 141 with its own internal price — see below.
Fuel consumed	0.62 million ℓ/y	232.0 million ℓ/y	The left is what the project buys. The right is not bought. Of it, 123.6 million would be for the heat alone.
Annual fuel cost	1.1 $m/y	405.9 $m/y	At 1.75 $/ℓ delivered.
Emissions	1,655 t CO_2/y	621,633 t CO_2/y	See Section 6.3.
Fuel logistics	One load per season	≈22 tankers a day	Along a route that floods from December to March.

The levelised cost of the self-supplied electricity has been calculated with the same method the model applies to heat — 8% discount rate, 25-year life, on the complete electrical capital including the reserve and the cleaning systems — and on the energy actually delivered to the load, not on that generated. It is the conservative comparison. Both diesel columns have been valued at the delivered price of emergency fuel; a bulk supply of that scale would have a different logistical structure and probably a different price per litre, but the margin is a factor of three and does not depend on that detail.

The comparison is the counterweight to the capital of Section 5.1. Two thirds of the energy this plant consumes is heat, and heat is produced here at 63 $/MWh-t against 206 for diesel. That is where the operating case is won, and it is why a system costing USD 1,107 million to build is nonetheless the cheap option over forty-five years.

TRACEABILITY

The model carries two diesel prices, and they are not in conflict

A reviewer opening the workbook will find 1.75 $/ℓ in one place and 1.20 $/ℓ in another. They correspond to different things. The 1.75 is the delivered price of the fuel the project *does* buy: small volumes, taken to 3,700 metres, to a 772 m³ tank farm. It moves real cash and it is the one this document uses in every comparison. The 1.20 lives in the solar thermal field tab and prices the fossil boiler island it was decided not to build.

The path of that cell has been verified: it feeds only the levelised cost of heat and the payback period that the same tab shows. No cell outside it reads it. The Dashboard takes from the solar thermal field its net capital cost, which does not depend on the fuel price. Consequently **the out-of-date price changes no capital, no operating cost, no debt service coverage and no return in this model**.

Were it restored to 1.75 $/ℓ, the annual saving attributed to the solar thermal field would go from USD 136.6 million to USD 149.6 million and the payback from 4.3 to 3.9 years. It is a presentational improvement and decides nothing; nor is it obvious that 1.75 is the correct substitute, because it is a delivered price for emergency volumes and not for a bulk supply of 124 million litres a year. It is left as it is, and noted here so that the coexistence of the two figures is not read as a contradiction.

5.4 Cell replacement

The battery does not last the forty-five years of the project and the model recognises this explicitly, which is unusual and correct. Lithium iron phosphate cells are limited by calendar and by cycles: at a rate of approximately one full cycle a day, in a heated enclosure at 3,700 metres, the planning life is fifteen years.

PARAMETER	VALUE	NOTE
Planning calendar life	15 years	12 would be a severe duty or deficient thermal management; 20 would be optimistic.
Replaceable fraction of the battery capital	75%	Cells, management system and thermal packs. Enclosures, HVAC, inverters, civil works and cabling are not replaced.
Installation uplift	1.25 ×	Removal, installation and commissioning. Below the 1.40 of new build: roads, platforms and connection already exist.
One replacement, in today's money	101.3 $m	On the plant battery.
Schedule	Years 20 and 35	Energised in year 5; cells are not bought with less than 5 years of operation remaining.
Total replacement, nominal, over the life	**199.2 $m**	**Charged to the cash flow available for debt service in the years indicated.**

Two observations of method, both conservative. The first: the replacement is treated as *additional* to the sustaining capital reserve of 5% of revenue, and not absorbed by it, even though that reserve amounts to USD 1,325 million over a fifteen-year cycle against a replacement of USD 101 million. It is the defensible presentation to a lender. The second: the replacement is not depreciated, so that no tax shield is taken against it; the real after-tax cost is between 25 and 32% lower than that shown.

5.5 What it means for the financing

The project's minimum debt service coverage is **1.163** and occurs in year 5, during the plant construction window. That is the point that sets the whole financial case, and the energy system touches it in three ways which should be declared separately.

> **What sets it.** Two things move this number. The cover of the UFK guarantee is at 90% — the standard product — which lowers the blended interest by 19 basis points but carries a premium of USD 44.7 million. Against that runs the capital: at a total project cost of USD 3,465.4 million the year-5 test gives **1.163**, and the solar thermal field is the largest single contributor to that capital.

- **The firm-supply reserve does not move the minimum coverage, and the reason is structural.** The gensets are financed within the captive energy sub-tranche, whose interest during construction is capitalised until commissioning and is paid from year 6. The charge therefore does not reach the year-5 test and the minimum coverage does not move. Were the same capital financed as ordinary plant capital, it would cost on the order of 0.013 of coverage. **This is an effect of financing structure, not a free lunch, and it is put to the credit insurer on that basis.** The figure to cite is 1.163; the figure to disclose is on the order of 1.150. *The 0.013 differential is cited as an order of magnitude.*
- **The backup runs 205.9 hours a year and carries no production.** It exists to hold the plant safe, not to generate. Section 4.5 sets out two measures that reduce those hours: committing in writing to forecast pre-charging takes the figure to 146.5 and costs no capital; four more hours of battery takes it to 161 and costs USD 42 million.
- **The operating cost is 4,867 $/t, and the energy system is 3.2% of it.** The solar thermal field's maintenance sits inside the general materials and spares line rather than a named one of its own.

It is also worth bearing in mind the comparison the model itself ran: safe hold beats doing nothing both in coverage and in return, even assuming a pessimistic hold load of 20 MW; and full production continuity costs 0.076 of coverage and 70 basis points of return in order to protect a production that the project can afford to lose. *Their direction is robust — safe hold protects the asset for one sixth of the cost of protecting production — and their magnitudes are indicative.* The sponsor's after-tax return is 11.9%.

SECTION 06

6. Non-energy consequences

A self-sufficient supply system of this size has consequences that are not electrical: it occupies land, consumes water to keep itself clean, avoids emissions and fuel logistics, and stores fuel within a sensitive basin. This section treats them in that order, and it is the one of most interest to the community reading of the document.

6.1 The land footprint

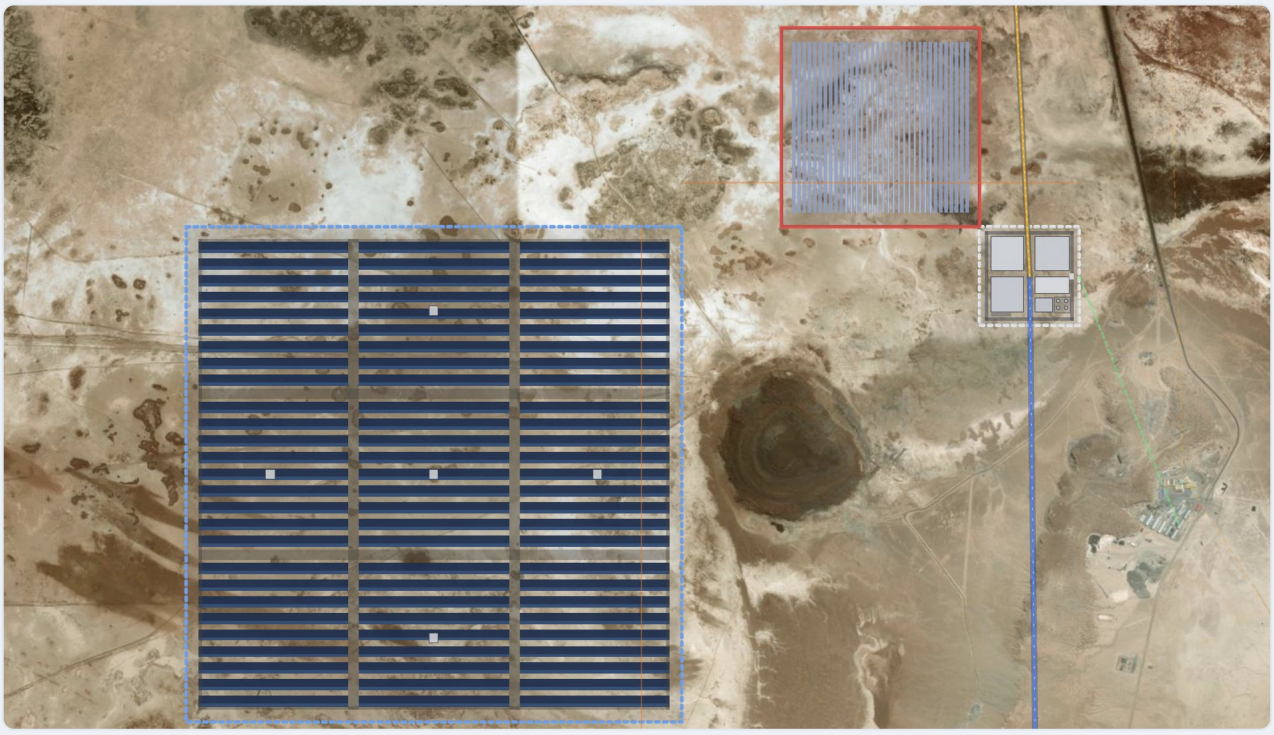


Figure 4 — Detail of the plant site according to the master plan: photovoltaic field to the west — 2.5 × 2.5 km — solar thermal field to the north — 1 × 1 km — and process plant to the east. The photovoltaic area allocated *amply accommodates* the 283.6 MW field of the current basis, at normal fixed-mount density and with a margin on the order of 32%. Satellite base: Esri / Maxar.

The master plan allocates the photovoltaic field a square of 2.5 × 2.5 km, that is 6.25 km², and the solar thermal field one of 1 × 1 km. With fixed-tilt structure, the density of a photovoltaic field lies between some 60 MW/km² for a normal layout and some 76 MW/km² for a dense layout. On 283.6 MW the required density is 45.4 MW/km²: **the field fits amply, and it fits below normal density.**

CRITERION	AREA REQUIRED	READING
Photovoltaic — dense layout, 76 MW/km²	3.73 km²	Occupies 60% of what is allocated; at that density the land would take up to 475 MW.
Photovoltaic — normal layout, 60 MW/km² *(base case)*	**4.73 km²**	**Fits within the 6.25 km² allocated with a margin of 32%. A conservative criterion and the one this document adopts.**

CRITERION	AREA REQUIRED	READING
Photovoltaic at P90 (21% capacity factor), 60 MW/km²	5.18 km²	Fits, with a margin of 21%.
Photovoltaic at 19% capacity factor, 60 MW/km²	5.72 km²	Fits, with a margin of 9%. Corresponds to no estimate; included to show the slope.
Total allocated by the master plan	**7.25 km²**	**6.25 km² of photovoltaics plus 1 km² of mirrors, separate.**

To size the scale in comprehensible terms: the 7.25 km² allocated are some 725 hectares, on the order of 0.07% of the area of the salar, on land adjacent to the plant and not on the crust of the interior.

6.2 The cleaning water

Every solar plant on a salar has the same problem: saline dust cannot be removed dry, it has to be washed off, and washing requires low-mineralisation water because hard water leaves deposits that reduce transmittance by more than the dust removed. Desert solar plants solve this by hauling water. This project does not.

The reason is structural and is developed in the accompanying document: the pond-free flowsheet cannot concentrate brine without producing distillate, and that distillate exceeds process demand. The cleaning of panels and mirrors is supplied from that surplus.

WATER BALANCE

The process produces more water than it consumes

The plant's water balance is set out in the Hydrological and Hydrogeological Assessment, Section 5, and this document takes it from there: the process produces more water than it consumes, with a **surplus of 8,165 m³ a day**. The committed supplies — 200 m³/day to Río Grande, 274 to the camp and 150 to panel washing, some 624 in total — take less than 8% of that surplus. The community commitment is amply covered.

The consequence should be stated in the right direction, because it is usually stated the other way round: this project's energy system does not represent a demand for water on the territory. It is a consumer of a by-product that would otherwise have to be disposed of.

And it is a difference from what already operates on the salar. Third-party sources note that the potash and lithium plants YLB operates today at Uyuni run below capacity for lack of water, and warn that water could cap production before energy does. That constraint does not carry over to this project for a reason of flowsheet and not of merit: the pond-free route evaporates in closed crystallisers and produces its own condensate, so that it generates water instead of consuming it. It is an argument worth bearing in mind when comparing this project with those already existing on the same salar, and it is developed in full in the accompanying hydrogeological assessment.

6.3 Emissions and fuel logistics

The relevant comparison is against diesel generation, which is what this supply replaces in practice.

ITEM	DIESEL EQUIVALENT	PROJECT	AVOIDED
Electricity — 433.4 GWh/y	290,378 t CO_2	—	290,378 t

ITEM	DIESEL EQUIVALENT	PROJECT	AVOIDED
Heat and steam — 1,050.6 GWh-t/y	331,255 t CO_2	—	331,255 t
Emergency backup	—	1,655 t CO_2	−1,655 t
Annual net	**621,633 t CO_2**	**1,655 t CO_2**	**619,978 t**
Over 45 project years			**≈ 27.9 Mt**

The figure the credit insurer should cite in its climate assessment is that of the solar thermal field net of the backup — **329,600 t CO_2 a year** — which is the one the model produces on an audited basis. The electrical figures in the table above are this document's arithmetic on the same emission factor, and serve to size the whole, not to replace that one. **The thermal side is where the climate case is won:** two thirds of the energy this plant consumes is heat, and producing it with sun instead of with diesel is what avoids most of the carbon.

NATIONAL CONTEXT

The fuel the project does not buy is the fuel the country does not have

The argument against a fossil supply is not only one of cost or emissions: it is one of availability. Bolivia has been a net importer of fuels since 2022 and the shortage of foreign currency makes it difficult to import diesel and LPG. During 2025 and 2026, in Potosí, residents queue — sometimes from before dawn — for LPG cylinders rationed to some hundred people a day at some stations; local transport declared an emergency for lack of diesel and petrol; and border areas such as Villazón reported a complete absence of supply.

A project that had chosen production continuity with diesel would have needed 6.15 million litres a year in that environment, and a captive gas plant would have pointed its demand at the resource that is the determining scarcity of the entire Bolivian energy system. The design adopted consumes 0.618 million litres a year in 206 hours, stored by season. **The advantage is not only environmental: it is that the project does not compete for a fuel that is rationed in the nearest city.**

The logistical consequence merits stating separately because it is the one perceived in the territory. An equivalent diesel supply would consume some 635 m³ of fuel a day, that is on the order of **twenty-two tankers a day**, entering and leaving along the same route for forty-five years. The model itself notes that this route floods between December and March and that a diesel fleet cannot guarantee transport all year round — that is, it fails precisely in the season in which the cloud events occur. The supply adopted requires filling a tank of eight hundred cubic metres once per season.

6.4 The stored fuel

The 772 m³ tank farm is the project's only permanent fossil infrastructure and it should be treated precisely, because it is where the environmental exposure of the energy chapter is concentrated. It carries fuel for the electrical deficit alone: the molten salt covers the thermal side of the event in full, so the backup boiler burns nothing.

- **Sizing.** It covers the whole wet season — four design events — with a 25% margin. It is filled before the season, precisely because the route is not reliable during it.
- **Containment.** Bunded tanks, impermeable lining and fire-fighting system. The costing also includes heating of tanks and lines and winterised fuel, because nights reach −10 °C and diesel gels.

- **Relative scale.** Under full production continuity the farm would have been 7,700 m³. Seven hundred and seventy cubic metres is a small bunded installation; eight thousand is a depot, and a depot of that scale within the Salar de Uyuni basin would have opened a schedule risk under the Ramsar Convention and in the environmental impact assessment. The safe hold decision closes that discussion before it begins.
- **What remains to be resolved.** The location of the farm relative to the shallow water table and to the seasonal flooding regime has not been examined, and it depends on the same behaviour of the crust that the hydrogeological document flags as unstudied. It must be resolved in site engineering, not in detailed engineering.

OPERATING PROCEDURE

What the tank farm does not have to cover

The fuel sizing rests on an assumption that can be improved without spending anything. The two measures of Section 4.5 — forecast pre-charging of the battery and mechanical panel clearing during the event — take the autonomy on battery alone from 44.5 to 130 hours if they are executed together *with the whole field cleared*, that is above the 96 hours of the design event. With half the field cleared, 81.5 hours are reached, below the event: the whole field is what the mitigation asks for.

Neither of the two is committed in writing today, and for that reason nothing has been deducted from the fuel or from the backup. They should be written down as an operating instruction — and audited as such — before commissioning. **This is the cheap route for reducing the running hours**: pre-charging on its own takes 205.9 hours to 146.5. Section 4.4.

6.5 The magnesium route, and why it is not yet settled

The flowsheet discharges residual magnesium as a bittern liquor amounting to 12% of the brine volume, which is then reinjected into a deep well. A saturation check added to the workbook on 9 September puts that liquor at **178 g/L of magnesium against a bischofite ceiling of 121 g/L** — 47% over-saturated. The two statements cannot both hold, and which one gives way has consequences for this document.

Either the bittern volume rises to **2.16 Mm³/y** — 18% of the brine instead of 12% — or some **2.2 Mt/y of $MgCl_2 \cdot 6H_2O$ crystallises as a solid**. On the second basis the solids to be moved rise from 3.67 to 5.9 Mt/y, which adds a further 0.8–1.4 MW to the haulage load declared in 2.1. On the first, water leaving as liquor is water not evaporated, so the condensate surplus of 6.2 is reduced by some 1,900 m³/day, to about **6,300 m³/day**. The committed supplies of 624 m³/day are met either way.

It is a permitting question before it is an energy question. Deep-well reinjection of concentrated magnesium chloride and a solid bischofite facility are two different environmental cases, with different studies and different consents. Bischofite is also deliquescent — it takes up water above roughly 33% relative humidity and dissolves — so on a salar with 350 mm of rain a year and a four-month wet season it cannot be treated as an inert stockpile. The route must be settled before the environmental submission, not after it.

SECTION 07

7. Results

NO.	RESULT
1	The site has no reliable electricity supply. Self-sufficiency is a physical constraint of the site, not a design choice or an environmental preference. The project takes no energy from the regional grid in any operating case, and there is no tariff at YLB cost for energy, water or brine.
2	Total energy demand is 169.4 MW — 49.5 MW electrical continuous and 119.9 MW-t thermal — equivalent to 1,484.0 GWh a year, or 18,550 kWh per tonne of carbonate. Two thirds of the thermal load is the steam that evaporates the brine. The intensity is 24% above the upper end of the range the literature cites for brine plants, and it is a direct consequence of the flowsheet: evaporating with steam reused 3.5 times costs five times as many units of energy as doing it with fully electric recompression.
3	The photovoltaic field is 283.6 MW with fixed-tilt structure, and that figure depends entirely on a capacity factor of 23% that has not been measured at the site. At P90 — 21% — the field grows to 310.6 MW, the capital rises by some USD 28 million and the area required goes to 5.18 km², which still fits amply within what is allocated.
4	Storage is 791.6 MWh with 16 hours of autonomy. That covers a night with margin; it does not cover a multi-day storm, and was never intended to.
5	Firm supply is resolved through safe hold and not through production continuity. Shedding 76% of the load, the storage already bought sustains 44.5 hours of electrical autonomy and 112.4 of thermal autonomy — the thermal event is covered in full. The diesel backup covers 51.5 of the 96 hours of the electrical event: 21.2 MW, USD 18.7 million.
5B	The binding design case is snow deposited on the panels, not cloud. With a residual output of 10% the field contributes 6.5 MW against a hold load of 12 MW, so that it covers a little more than half. Mechanical panel clearing remains a first-order mitigation measure, not formalised today, but with the smaller field it delivers less: the whole field has to be cleared, and not half of it, to close the event without diesel.
6	The energy system costs USD 1,106.7 million of capital — 32% of the total project cost of USD 3,465.4 million — and USD 157.2 per tonne in named line items, 3.2% of the operating cost. This project's energy is expensive to build and cheap to operate, and that is the shape of risk actually being financed.
7	The supply adopted costs a third of diesel generation, both in electricity — 139 against 438 $/MWh, the latter fuel only — and in heat — 63 against 206 $/MWh-t. Two thirds of the energy the plant consumes is heat, which is the cheaper of the two by a factor of three.
8	The project avoids on the order of 619,978 t of CO_2 a year relative to an equivalent diesel supply, some 27.9 million tonnes over the life of the project, and it also avoids the traffic of some twenty-two tankers a day along a route that floods a third of the year.
9	The land allocated is sufficient. The 283.6 MW photovoltaic field requires 4.73 km² at normal density and the master plan allocates 6.25 km²: a margin of 32%, which also absorbs the P90 case. The solar thermal field has a 1 km² enclosure of its own.
10	The cleaning water — 18 to 24 washes a year, with low-mineralisation water — comes from the surplus condensate of the process itself. The energy system does not represent a demand for water on the territory; it is a consumer of a by-product.
12	The minimum debt service coverage is 1.163 in year 5. The firm-supply reserve does not move it, because it is financed within the captive sub-tranche whose interest is capitalised beyond the year-5 test; financed as ordinary plant capital it would cost on the order of 0.013 of coverage.
13	The backup runs **205.9 hours a year** and carries no production in any operating case. The figure follows from a 792 MWh fleet against a 12 MW hold load over a 96-hour event.
14	Wind has not been measured at the site, and the available literature points against it: an annual mean of some 5.0 m/s at 100 m, air density at 69% of that at sea level, and — decisively — a high-wind period from June to October against a design event from January to March. Microreactors were assessed and discarded as a design option, but they remain alive as a repowering option in the year-20 replacement decision.

SECTION 08

8. What must be confirmed, and in what order

The results above fall into three groups: those that can be closed at the desk this week, those that require a season of measurements, and those that require vendor commitment and detailed engineering. They should be addressed in that order, because the desk items cost nothing and change the questions the expensive ones have to answer.

8.1 Can be closed now, with no field work

The first two are questions for the process engineer and cost nothing. The rest are enquiries that can be sent this week.

- **The salt yields and the magnesium route at the design grade** — tonnes of each salt per tonne of carbonate at 1.5 g/L, and whether the residual magnesium leaves as liquor or as solid bischofite. One request to the process partner, from a crystallisation model that already exists. It closes both new rows of 9.3, the haulage load of 2.1 and the environmental case of 6.5.
- **The basis of the indicative cost for solar steam with a vacuum crystalliser** — whether it is annualised capital, operating cost or full cost per tonne, at what scale, and whether the solar field is included. 5.1 identifies the USD 586.7 million solar thermal field as the largest and least verified item in the system; this is the first external figure against which it can be tested.
- **The actual hourly profile of the electrical load** — shape over the day, peak-to-mean ratio, and what fraction is sheddable. The size of the photovoltaic field and that of the batteries depend on it simultaneously, and together they are on the order of USD 514 million of the system's USD 1,107 million. Today an average power is treated as though it were constant.
- **The genuine hold load and the hold temperature**, plus how many hours a controlled drain-down and shutdown actually takes. The whole of Section 4 scales on those figures, and so do the annual running hours of the backup.
- **Obtain a quotation for the specific consumption of the recompression compressor** from Protarget or Ebner. The model's 10 kWh/m³ is a declared placeholder; the plausible range runs from 0 to 35, and the high end takes the project's electrical load to 60.7 MW with everything that drags along. Section 2.2.
- **The per-stage thermal intensities** against the mass and energy balance of the carbonate flowsheet. The gross heat figure governs a third of the thermal load, and with it the aperture, the capital and the levelised cost of the solar thermal field.
- **Enquiry to the insurer** as to whether a firm safe hold capacity reduces the business interruption premium. That credit is not modelled and can only improve the case.
- **Commit in writing to the pre-charging procedure** by forecast and to that for mechanical panel clearing, with the responsible person, the trigger threshold and the priority clearing block defined in advance. They are the two mitigations that cost no capital, and together they take the autonomy on battery alone from 44.5 to 130 hours — above the 96 hours of the design event — with the whole field cleared. Section 4.5.

8.2 A season of measurements

- **A solar measurement station at the site** — global and direct normal irradiance, with pyranometer and pyrheliometer — installed at the earliest possible date, given that a full year is the minimum useful record. It is the measurement that converts both the photovoltaic capacity factor and the direct normal irradiance from literature values into data. The second weighs more than the first: it sets the aperture of the solar thermal field directly — the mirror field, whose solar multiple was raised until the annual energy check closed at zero — and that field is the largest capital item in the system.
- **Satellite record and typical meteorological year** for Uyuni: the actual frequency and duration of episodes with irradiance below 2 kWh/m² a day between January and March, and how often the snow actually settles on the panels rather than merely clouding over. It is what turns the design event from a statement into a measurement, and it can be started immediately on records that already exist.
- **Wind resource screening** by met mast or lidar, framed as an exclusion exercise and not as development. The question it must answer is a specific one: what wind there is *during* the episodes of irradiance below 2 kWh/m² a day between January and March. The annual mean is already known regionally and it is poor; if the measurement confirms that there is no wind during the events either, wind is closed off and stops consuming attention. It is not incorporated into any figure in this document.
- **Soiling rate** measured on test coupons exposed outdoors. It establishes the cleaning frequency, the water consumption and the cleaning cost on a measured rather than a scaled basis.
- **Temperature record** at panel height and at battery enclosure height, through a complete winter. It feeds the derating curve of the gensets and the parasitic draw of the cell heating.

8.3 Requires vendor commitment and detailed engineering

- **Contractor quotation** for the photovoltaic field at 3,700 m. The 0.73 $/W direct — on the order of 1.10 $/W installed — is a planning figure built by analogy, not an offer.
- **Turnkey price for the battery system** explicitly covering active cell heating and liquid thermal management with insulation, not cooling alone. It must appear in the specification in those words: it is the difference between a system that works at −10 °C and one that will not accept charge.
- **Certified derating curve** for the gensets at 3,700 m and −10 °C, and confirmation of black start from a discharged battery, which is the real condition in which they have to start.
- **Quotation and validation of the solar thermal field with the designer.** It is the largest item in the system — USD 586.7 million — and the least verified: 250 $/m² of collector field and 35 $/kWh-t of storage are planning values, not offers. It must also be confirmed that the field scales linearly or better — this project is on the order of four times the tonnage of the reference installation and 1.14 times its aperture — and what the ground layout of the proposed collector is.
- **Contracting structure for the field.** The field is a single one; what is open is whether it is tendered as one 283.6 MW package or as separate packages by block. A single tender captures economies of scale in civil works, medium-voltage architecture, control room and operating contract. The financial model keeps the blocks separate for cost allocation reasons, which must not be read as a construction decision.

SECTION 09

9. Basis and sources

9.1 Project sources used

Every quantitative figure in this document comes from the project's techno-economic model in its v22 version as at 28 August 2026, or is derived arithmetically from it. The tabs used, and what was taken from each, are as follows. **Where a tab retains an out-of-date label or table, this is expressly stated in the right-hand column and in the corresponding section.**

SOURCE	WHAT WAS TAKEN
Evaporation basis	Evaporation basis: 49 t of water per tonne of carbonate, steam reuse 3.5×, and the consumption of the recompression compressor. It heads the list of as-yet unverified assumptions collected in Section 9.3.
Power	Electrical load of the plant and auxiliaries, capacity factor, photovoltaic and battery sizing, capital build-up, reserve margins, cleaning systems and cost, cell replacement schedule, and the pilot phase portion. It is the principal tab for this document.
Notes	History of design decisions: elimination of the grid mode, adoption of safe hold, correction of the capacity factor, fixed-tilt structure decision, cell heating, land footprint, and the assessment and discarding of microreactors. The warning at the head of the tab has been heeded: where a note differs from a live figure, the live figure was used.
Firm Supply (Diesel)	Design event, hold load, autonomy on battery and on salt, deficit, sizing and cost of the gensets, fuel, running hours and tank farm. **The autonomy column of the sensitivity table in this tab is calculated on a 1,063.9 MWh battery fleet** rather than the 791.6 MWh installed; the autonomy, operation and hours columns of Section 4.6 are recalculated here on the installed fleet.
Heat (CSP)	Gross and net thermal loads, evaporation steam, solar multiple, aperture, annual balance check, salt storage, capital build-up, levelised cost of heat and avoided emissions. It is today the tab of greatest weight in the document. It does not calculate the land area of the field.
GEA & Grade	Volume to be evaporated, latent heat and steam reuse factor, specific consumption and load of the recompression compressor, associated photovoltaic field and battery and their unit costs, and the process water balance with the condensate surplus. It is the current source of those figures.
Opex Build-Up	Structure of the bottom-up rebuilt operating cost, and the insurance and materials items that the model charges automatically on the plant capital. Also the 702 permanent positions and the community programmes of the “The project” page.
Dashboard	Total project cost, plant capital, total operating cost, debt service coverage, sponsor's return, and the lock-out of the supply mode. Also the royalty regime switch of the “The project” page. **It is the source that governs where another tab differs**, under the criterion adopted in this revision.
Returns · Camp & Community Water	Only for the “The project” page: the 15% community reinvestment on distributions, and the population of the residential camp. Neither feeds any energy figure in this document.
Water Recovery	Annual design irradiance. The water circuit sizing contained in this tab **is inactive**: it does not feed the model and has not been used here. The current source is GEA & Grade; see Section 2.2.
Site master plan	Layout and areas allocated to the photovoltaic field, the solar thermal field and the process plant. Basis of Figure 4.
Hydrological and Hydrogeological Assessment v1.4	Water supply commitments, for Section 6.2. Accompanying document: **its condensate surplus of 8,165 m³/day is the reference water balance for this document**. Section 6.2.

9.2 Third-party sources cited by the model

The following are not sources of this document but of the model from which it is taken. They are listed so that they can be verified independently, and none has been checked here against its original publication.

- Performance study by an independent consultant for a 300 MWp photovoltaic project on the Altiplano, which uses 2,050 kWh/kWp as maximum yield, equivalent to a capacity factor of 23.4%.
- The solar thermal energy installation of the Zabuye lithium plant, in Tibet, with 621,300 m² of collector field, used as a scale-up reference.
- Industrial references for mechanical vapour recompression on high-salinity brines, for the specific consumption of 35 kWh/m³ and the evaporator unit cost.
- Market prices for lithium iron phosphate battery systems: 110 to 125 $/kWh in markets with a supply chain, and 180 to 260 $/kWh at remote sites.
- Cost range for molten-salt thermal storage of 25 to 70 $/kWh-t, from the German solar thermal industry association.
- Removal of fuel subsidies in Bolivia in December 2025, with pump diesel going from 3.72 to 9.80 bolivianos per litre — on the order of 1.42 $/ℓ in February 2026, a 163% year-on-year increase — before transport to 3,700 m.
- *Wind Power Potential in Highlands of the Bolivian Andes: A Numerical Approach* (Energies, 2022) — annual mean of some 5.0 m/s at 100 m for Uyuni, monthly range of 4.3 to 5.6 m/s, high-wind threshold of 7.5 m/s, high-wind period from June to October and daily maximum between 20:00 and 22:00. It is a numerical study of regional scale and does not substitute for a measurement at the site. Sole source of the wind figures of Section 1.3; it does not come from the project model.
- *Potosí, Bolivia: situación del gas y la electricidad y la demanda energética de una planta de litio* (Potosí, Bolivia: the gas and electricity situation and the energy demand of a lithium plant — energy report and feasibility note, July 2026) — SIN figures attributed to the digital yearbook of the AETN and the CNDC for 2024: 3,787 MW installed, 3,318 MW effective, 1,752 MW peak demand, 11,689 GWh of annual generation, 65 to 70% gas share. Also the fall of more than 56% in domestic gas production relative to 2015, the shortages of LPG and liquid fuels in Potosí and Villazón during 2025-2026, the 60 MW Uyuni solar plant of ENDE Guaracachi, the energy intensity range of 5 to 15 MWh per tonne, and the water constraint on the YLB plants in operation. **It is a secondary compilation**: its figures have not been verified against the primary sources in this assessment and the note itself warns that they must be confirmed before deciding. Basis of Sections 1.1, 2.4, 3.1, 6.2 and 6.3.
- International Standard Atmosphere for air density at 3,700 m — 0.845 kg/m³, 69% of the sea-level value. Applied to the wind derating of Section 1.3.

9.3 Declared assumptions and inconsistencies

The following do not come from the model, or come from it with a reservation. They are declared so that they can be challenged and, where appropriate, replaced.

ITEM	VALUE USED	WHY, AND WHAT WOULD REPLACE IT
Solar capacity factor	23%	From an analogous project on the same Altiplano, not from the site. The full range from 19 to 27% is shown in Section 1.2. It is replaced by a solar measurement station with a year of record.

ITEM	VALUE USED	WHY, AND WHAT WOULD REPLACE IT
Photovoltaic field density	60 MW/km²	Normal layout with fixed-tilt structure. The model also contains 76 MW/km²; the conservative one was adopted because the risk of undersizing the land is asymmetric. Compared in Section 6.1 with the 6.25 km² the master plan allocates. It is replaced by the contractor's layout design.
Safe hold load	12 MW	Engineering estimate of what must keep running. It is not a process balance figure. Section 4.6 runs from 8 to 25 MW. It is replaced by the process engineer.
Duration and frequency of the event	4 days, 4 per season	Sponsor's statement, not a meteorological record. It determines the fuel and the tank, not the installed capacity. It is replaced by the typical meteorological year.
Residual output during the event	0%	Snow on the glass. Adopted as the binding case; at 10% the field contributes 6.5 MW and the deficit falls to 5.5 MW, not to zero. It is replaced by the record of snowfall against cloud.
Cleaning frequency	18 to 24 a year	Estimate. The soiling rate has not been measured and the volume of water per cleaning is not yet documented in any source. It is replaced by test coupons exposed outdoors.
Altitude derating	72% of ISO rating	Handbook value for turbocharged gensets at 3,700 m. It is replaced by the vendor-certified curve at −10 °C.
Calendar life of the cells	15 years	Planning value. 12 corresponds to a severe duty, 20 would be optimistic. Capacity loss between replacements is not modelled.
Real fall in cell cost	2.5%/year	Deliberately conservative: the realised fall between 2015 and 2025 was on the order of 8% a year in real terms. Net of inflation it gives a nominal factor of approximately 1.00.
Installed cost of the panel	0.73 $/W direct	On the order of 1.10 $/W installed. Built by analogy for a remote, high-altitude site with no local supply chain. It is replaced by a contractor quotation. It is not the largest swing factor in the capital; the solar thermal field is.
Diesel price in the thermal calculation	1.20 $/ℓ (counterfactual)	The model's figure is retained. It prices a boiler island that is not built and feeds no financial figure — only the levelised cost and the payback that tab shows. The fuel the project does buy is the emergency fuel, at 1.75 $/ℓ. Section 5.3.
Design direct normal irradiance	**2,700 kWh/m²/y**	**Literature value for the Altiplano, within a band of 2,600 to 2,900. Not measured at the site. It sets the aperture of the solar thermal field, which is the largest item in the system and whose annual energy check closes at zero: at 2,600 the required aperture rises to 734,700 m². It is replaced by a station with a pyrheliometer. It is today the most influential assumption in the document.**
Consumption of the recompression compressor	**10 kWh-e/m³**	**A placeholder declared by the model itself, pending a quotation from Protarget or Ebner. The plausible range runs from 0 — pure multiple effect, with no compressor — to 35 — fully electric recompression, with no steam; at the high end the project's electrical load rises to 60.7 MW. Section 2.2.**

ITEM	VALUE USED	WHY, AND WHAT WOULD REPLACE IT
Water evaporated per tonne of product	**49 t/t**	**2026 technical review of the evaporation design: 30 t up to the solubility limit of lithium sulfate and the potassium salts, plus 19 t in the Route C stage. It is an editable cell in the model. It sets the energy load of the evaporation (Section 2.2); the water balance is taken from the Hydrological and Hydrogeological Assessment.**
Steam reuse	3.5 ×	Midpoint of the “3 to 4” of the technical review of the evaporation design. At 3.0 the peak of the solar thermal field rises to 432.5 MW-t; at 4.0 it falls to 356.6. It is replaced by the evaporator design. *The sensitivity note in the tab itself is inverted and has not been used.*
Operation and maintenance of the solar thermal field	**No line item of its own**	**The operating cost absorbs it within a generic 1.2% line on the capital not belonging to the process island, which grants it some 81 $/t. The field's own tab assumes 2% of its capital — 147 $/t — in calculating the levelised cost of heat. The difference, some 65 $/t, is not in the model's operating cost. Section 5.2.**
Evaporation sizing	Current basis	The current basis was used — 3.92 Mm³, solar steam at 627 kWh-t/m³ with 3.5× reuse, compressor at 10 kWh-e/m³. The model retains two inactive sizings: that of the *Water Recovery* tab — 3.92 Mm³, 30 kWh/m³, 27% capacity factor, 49.7 MW — and an all-electric train on 5.38 Mm³ at 35 kWh/m³, 93.5 MW. They are recorded here for traceability.
Parasitic consumption of cell heating	**Not disaggregated**	**It is known to exist and known to be small; it is quantified nowhere in the model and not deducted from the available energy.**
Salt load per m³ of brine	**302.5 kg/m³**	**Taken from an 85 t / 400 m³ basis which, worked back, corresponds to a brine of about 0.47 g/L lithium — the salar average — against a design grade of 1.5 g/L. Comparative salt-load data for Uyuni brines at other lithium grades imply 405 kg/m³ at 0.6 g/L rising to 581 kg/m³ at 2.6 g/L. A uniform recalibration was tested on 9 September and reverted: it drove sodium chloride to 343 kg/m³ against a halite-saturation ceiling of 237–259 kg/m³, which is not physically possible. The finding is therefore carried as a range — salt load 1.0 to 1.86× — and not adopted as a point. It scales the haulage load of 2.1. It is replaced by a wellfield assay and a thermodynamic crystallisation model of the actual brine.**
Magnesium disposal route	**Bittern liquor**	**The model routes residual magnesium to reinjection as a liquor. The saturation check says that liquor is over-saturated and that bischofite would crystallise. Worth 0.8–1.4 MW of haulage load and a different environmental case. See 6.5.**
Useful life for the levelised cost	25 years, 8% discount rate	The model's own method, applied here to electricity as well so that both figures are comparable. The project models 45 years of operation.

Observations

This assessment is delivered to be examined, and it is delivered to two readers with different interests without changing a single figure between one and the other. Observations from the technical advisers the addressees appoint may be addressed in writing to Heitaria Swiss AG, Rigistrasse 1, 6374 Buochs, Switzerland, and will be answered in writing.

Where an observation bears on a parameter, Section 9.3 identifies which ones are open to replacement and what measurement would close them. The arithmetic of Sections 2, 4 and 5 is set out in detail precisely so that another value can be substituted and the consequence followed without redoing the document.

ANNEX A

What European battery buyers will ask about energy and carbon

The EU Batteries Regulation (EU) 2023/1542 reaches this plant through its customers in two ways. Article 7 requires a carbon-footprint declaration for electric-vehicle, light-transport and rechargeable industrial batteries, calculated over the battery's life cycle with the method the Commission sets in delegated acts; the battery maker will ask its raw-material suppliers for their own data rather than use generic values. And from 18 August 2027 the supply-chain due diligence of Articles 47–53 lists energy use and air among the environmental risks the buyer must examine (Annex X).

This annex adds nothing to the energy analysis. It shows which of the buyer's questions this assessment already answers, and which belong to studies that do not yet exist. The per-tonne figures below are this annex's arithmetic on the document's own numbers at 80,000 t/y.

A.1 The questions, and where the answers are

WHAT THE BUYER WILL ASK	WHAT THIS ASSESSMENT ESTABLISHES	STATUS AND WHERE
Where does the plant's energy come from?	From a 283.6 MW photovoltaic field with 792 MWh of batteries and a 389.1 MW-t solar thermal field with molten-salt storage. No grid connection. Total demand 169.4 MW, about 18,550 kWh per tonne of carbonate, two thirds of it heat.	**Established** as architecture; capacity factor and hourly load not yet measured. §1–3, §8.
Does the plant burn fossil fuel?	Only for emergency safe hold: seven 3.5 MW diesel gensets, about 206 hours a year, 0.618 million litres, about 1,655 t CO_2 a year. The gensets protect the plant, not production; the backup boiler burns nothing.	**Estimate.** Running hours depend on snowfall frequency, for which no site record exists. §3.4, §4, §6.3.
What are the on-site energy emissions per tonne?	About 21 kg CO_2 per tonne of carbonate from the backup diesel. Producing the same energy with diesel would emit about 7.8 t CO_2 per tonne.	**Arithmetic** on §6.3. Covers on-site energy only.
What is the carbon footprint of the product?	Not calculated. A footprint to the EU method also counts what this document does not: production and transport of reagents, transport of the product and of fuel, and the manufacture of the solar field, batteries and plant.	**Open.** Requires a life-cycle assessment to the Commission's method.
Is there a risk of fuel spills?	A 772 m³ bunded tank farm with lining and fire protection, filled once per season. Its position relative to the shallow water table and the seasonal flooding has not been examined.	**Open** on siting. §6.4.

FOR EXTERNAL USE

What can be said today, and what cannot

What can be said: production is powered by the sun, with no grid and no fuel except an emergency reserve, and on-site energy emissions are of the order of tens of kilograms of CO_2 per tonne. What cannot yet be said: a carbon footprint for the carbonate. That figure should not be published until the life-cycle assessment exists, because a buyer will compare it against the EU method, not against diesel.